Set 4B: BJT Biasing with Current Mirrors

ECEN 325 — Lecture 4 (Bipolar Junction Transistors) practice set

What this set trains. DC analysis when the bias currents come from current mirrors. The reference current is set by a resistor and diode-connected transistor(s); every transistor that shares the base and emitter nodes of a diode-connected transistor copies its current. Then continue node by node as in set 4A and verify the assumptions. $\beta = 100$ throughout. Problems 1–6 are easy, 7–12 medium, 13–18 hard. Try each problem before looking at its solution.

The Lecture-4 pipeline (use it for every problem in this lecture)

Tools — this is all the theory you need (plus the Lecture-1–3 tools):

T1 *Active BJT at DC:* npn $V_{BE} = 0.7\,\text{V}$, pnp $V_{EB} = 0.7\,\text{V}$; $I_C = \beta I_B$, $I_E = (\beta+1)I_B$, so $I_C \approx I_E$ and, when β is large, $I_B \approx 0$. Active needs $|V_{CE}| \geq 0.2\,\text{V}$.

T2 *Current mirror:* transistors with the same base node and the same emitter node have the same V_{BE}, hence the same I_C.

T3 *Small-signal parameters:* $r_e = \frac{V_T}{I_C}$ ($V_T = 25\,\text{mV}$), $\alpha = \frac{\beta}{\beta+1}$, $r_o = \infty$. For AC, DC supplies are ground and "∞" capacitors are shorts.

T4 *Node impedances and gains:* $Z_{base} = (\beta+1)(r_e + Z_E)$, $Z_{emitter} = r_e + \frac{Z_B}{\beta+1}$, $Z_{collector} = \infty$; CE $\frac{v_c}{v_b} = -\alpha\frac{Z_C}{r_e+Z_E}$, CB $\frac{v_c}{v_e} = \alpha\frac{Z_C}{Z_{emitter}}$, CC $\frac{v_e}{v_b} = \frac{Z_E}{r_e+Z_E}$. Z_C, Z_E, Z_B are everything else hanging on that node (including the input resistance of the next stage).

Steps:

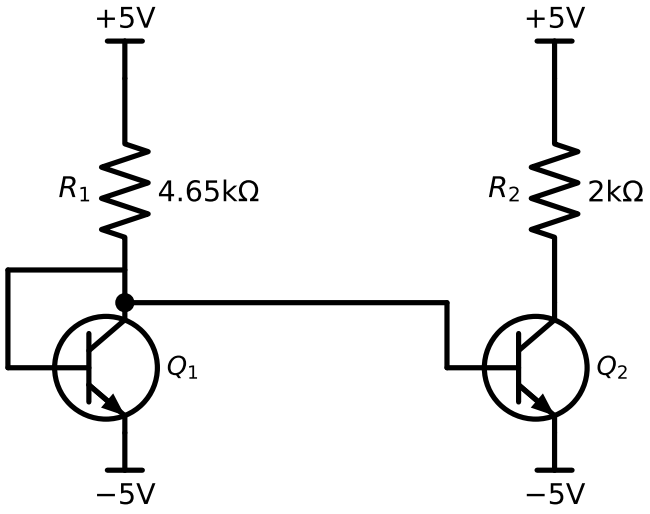
1. *DC:* assume every transistor active and $I_B \approx 0$; walk through the circuit node by node (dividers, V_{BE}, KCL for I_E, $I_C \approx I_E$, collector voltages); then *verify*: $I_B \ll$ bias currents and $|V_{CE}| \geq 0.2\,\text{V}$.
2. *AC:* r_e of every transistor; input resistance of every stage from the last one backwards; multiply the node-to-node gains along the signal path; then standard form, Bode plot and output as in Lecture 1.
3. *Design:* run the same relations backwards, choosing the free quantities with the stated design rules, then check the result by analysing the finished circuit.

Problems

Easy problems

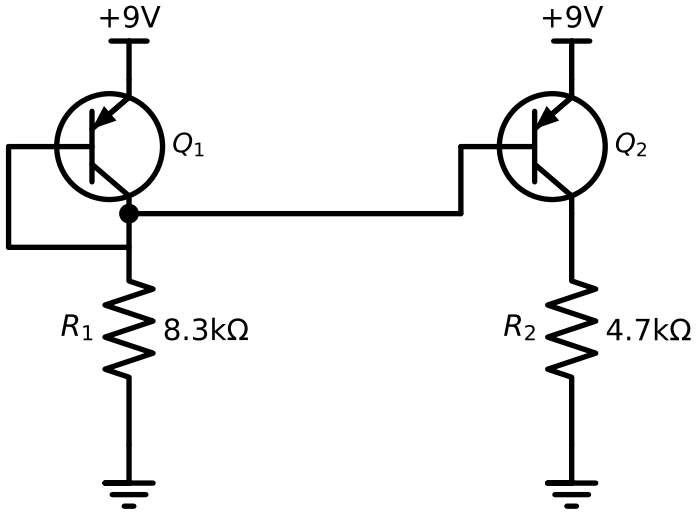
Problem 4B.1 **Easy**

For the circuit below, find the DC collector currents I_C of all transistors and verify all assumptions, including the operating region of every transistor. Assume $\beta = 100$.



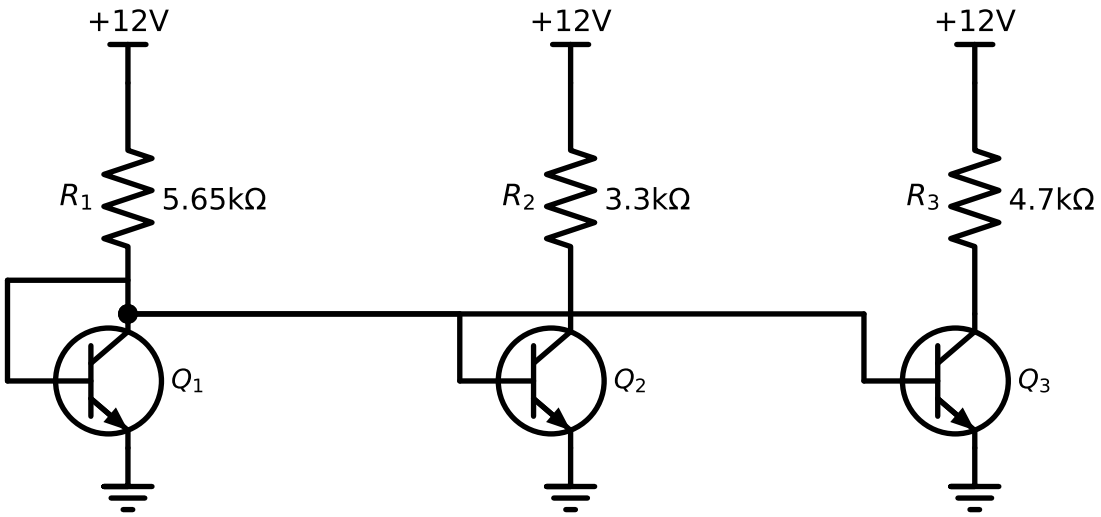
Problem 4B.2 **Easy**

For the circuit below, find the DC collector currents I_C of all transistors and verify all assumptions, including the operating region of every transistor. Assume $\beta = 100$.



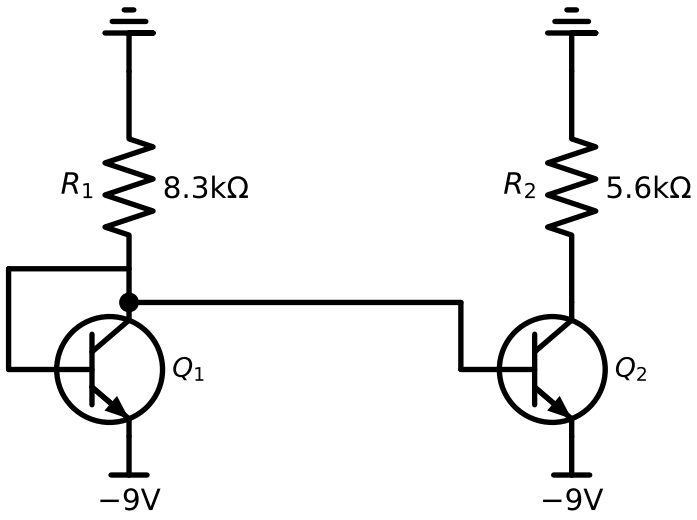
Problem 4B.3 **Easy**

For the circuit below, find the DC collector currents I_C of all transistors and verify all assumptions, including the operating region of every transistor. Assume $\beta = 100$.



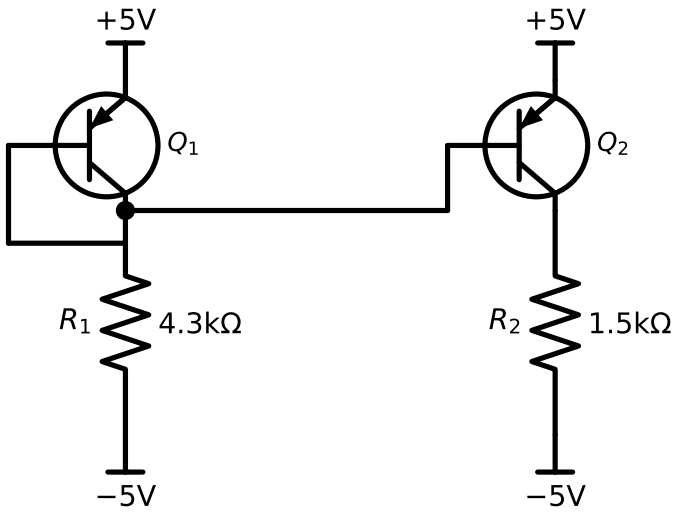
Problem 4B.4 **Easy**

For the circuit below, find the DC collector currents I_C of all transistors and verify all assumptions, including the operating region of every transistor. Assume $\beta = 100$.



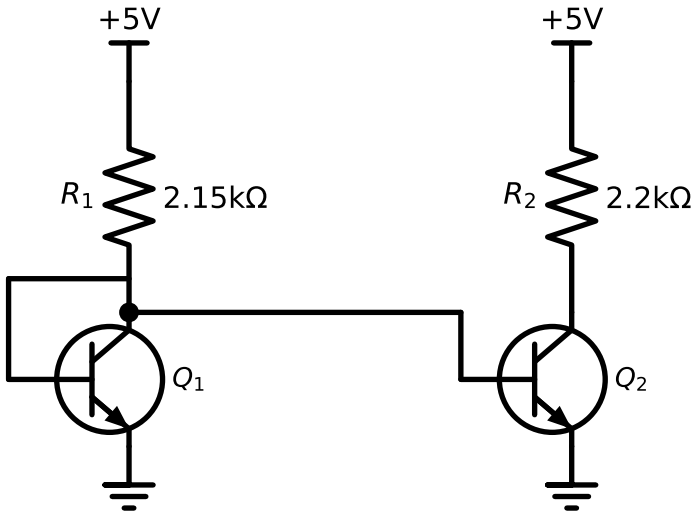
Problem 4B.5 **Easy**

For the circuit below, find the DC collector currents I_C of all transistors and verify all assumptions, including the operating region of every transistor. Assume $\beta = 100$.



Problem 4B.6 **Easy**

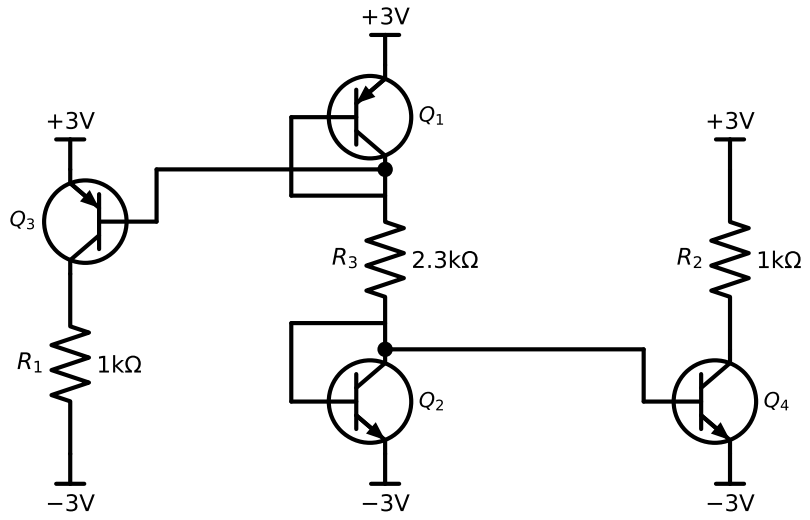
For the circuit below, find the DC collector currents I_C of all transistors and verify all assumptions, including the operating region of every transistor. Assume $\beta = 100$.



Medium problems

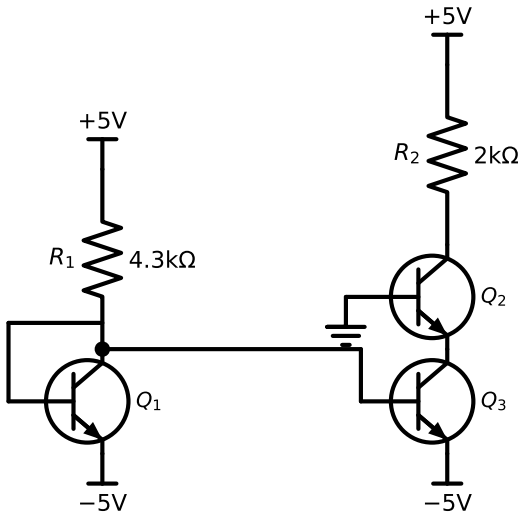
Problem 4B.7 — **Medium**

For the circuit below, find the DC collector currents I_C of all transistors and verify all assumptions, including the operating region of every transistor. Assume $\beta = 100$.



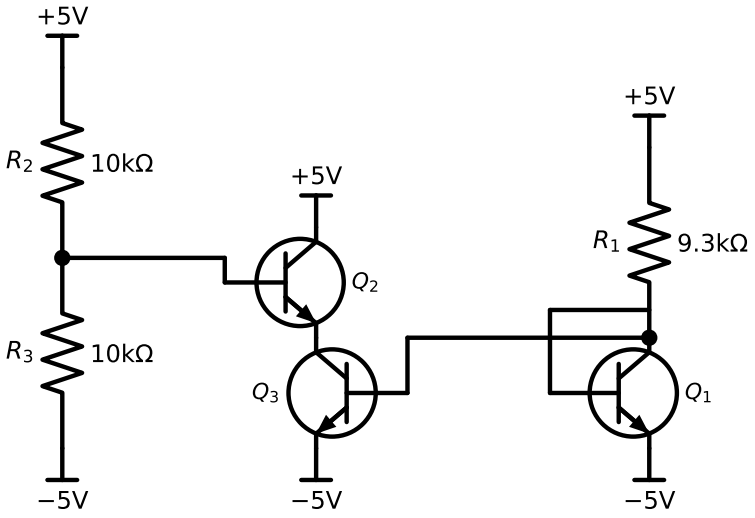
Problem 4B.8 — **Medium**

For the circuit below, find the DC collector currents I_C of all transistors and verify all assumptions, including the operating region of every transistor. Assume $\beta = 100$.



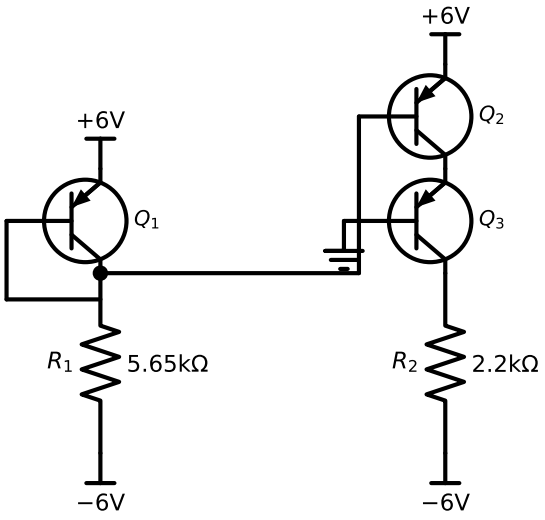
Problem 4B.9 **Medium**

For the circuit below, find the DC collector currents I_C of all transistors and verify all assumptions, including the operating region of every transistor. Assume $\beta = 100$.



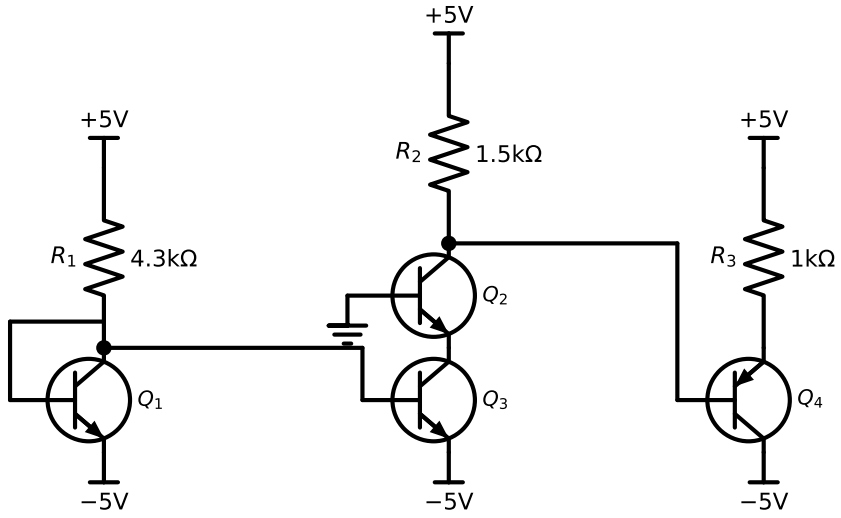
Problem 4B.10 **Medium**

For the circuit below, find the DC collector currents I_C of all transistors and verify all assumptions, including the operating region of every transistor. Assume $\beta = 100$.



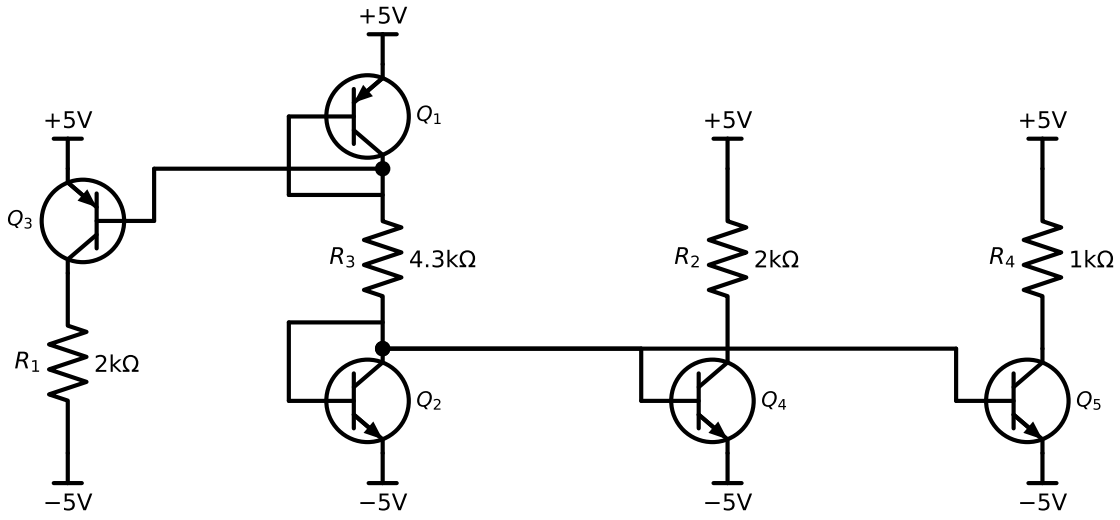
Problem 4B.11 **Medium**

For the circuit below, find the DC collector currents I_C of all transistors and verify all assumptions, including the operating region of every transistor. Assume $\beta = 100$.



Problem 4B.12 **Medium**

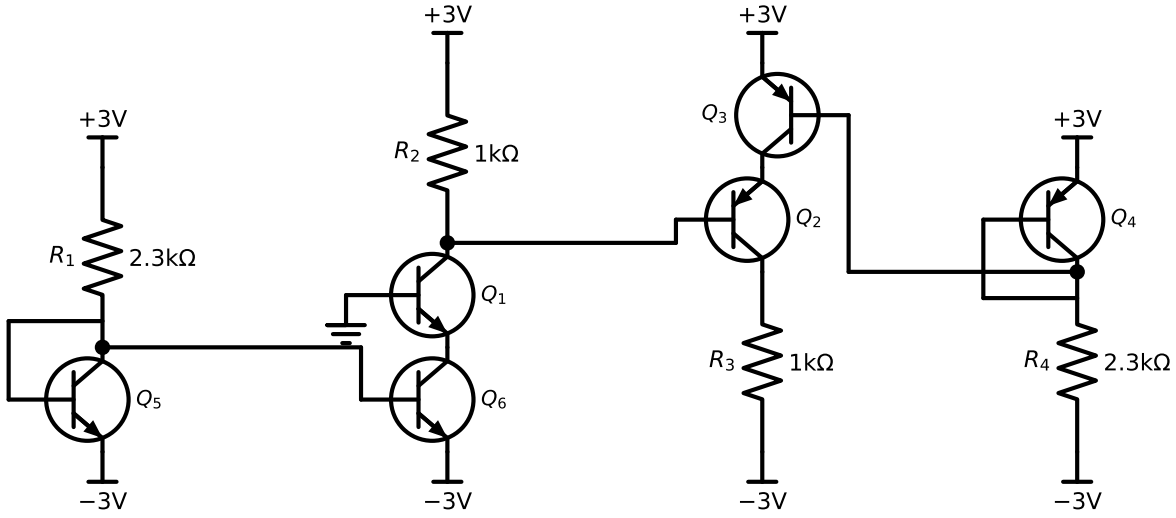
For the circuit below, find the DC collector currents I_C of all transistors and verify all assumptions, including the operating region of every transistor. Assume $\beta = 100$.



Hard problems

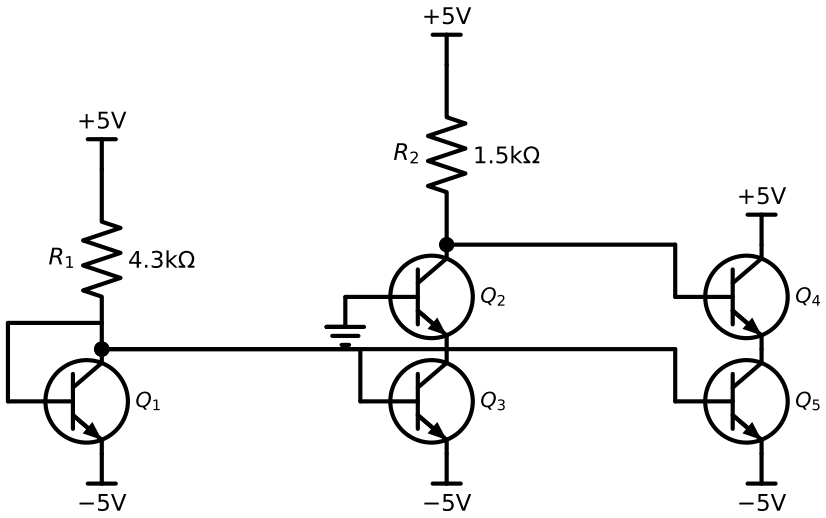
Problem 4B.13 **Hard**

For the circuit below, find the DC collector currents I_C of all transistors and verify all assumptions, including the operating region of every transistor. Assume $\beta = 100$.



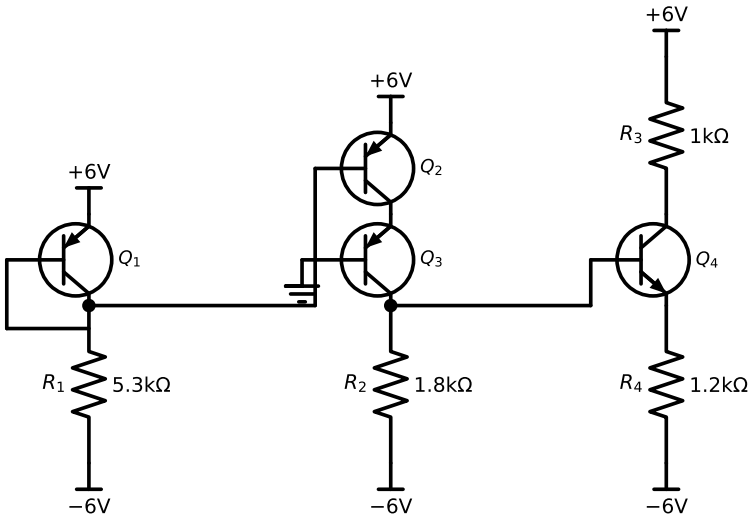
Problem 4B.14 **Hard**

For the circuit below, find the DC collector currents I_C of all transistors and verify all assumptions, including the operating region of every transistor. Assume $\beta = 100$.



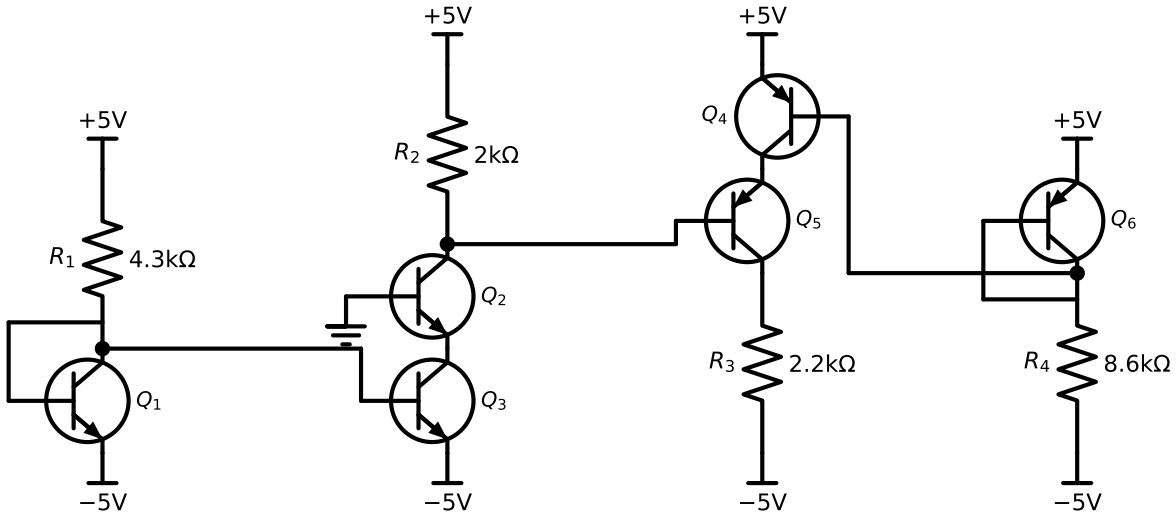
Problem 4B.15 — Hard

For the circuit below, find the DC collector currents I_C of all transistors and verify all assumptions, including the operating region of every transistor. Assume $\beta = 100$.



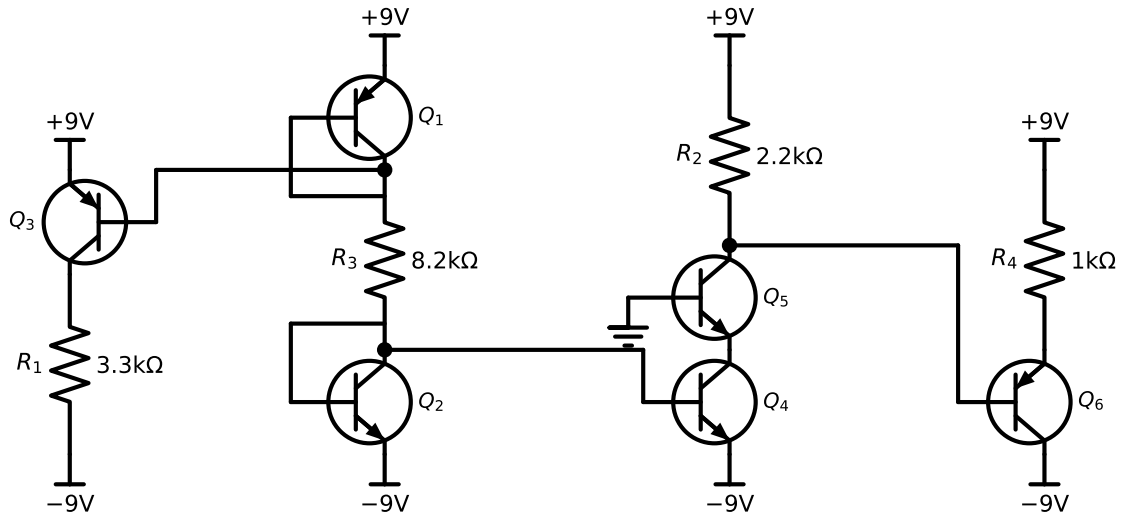
Problem 4B.16 — Hard

For the circuit below, find the DC collector currents I_C of all transistors and verify all assumptions, including the operating region of every transistor. Assume $\beta = 100$.



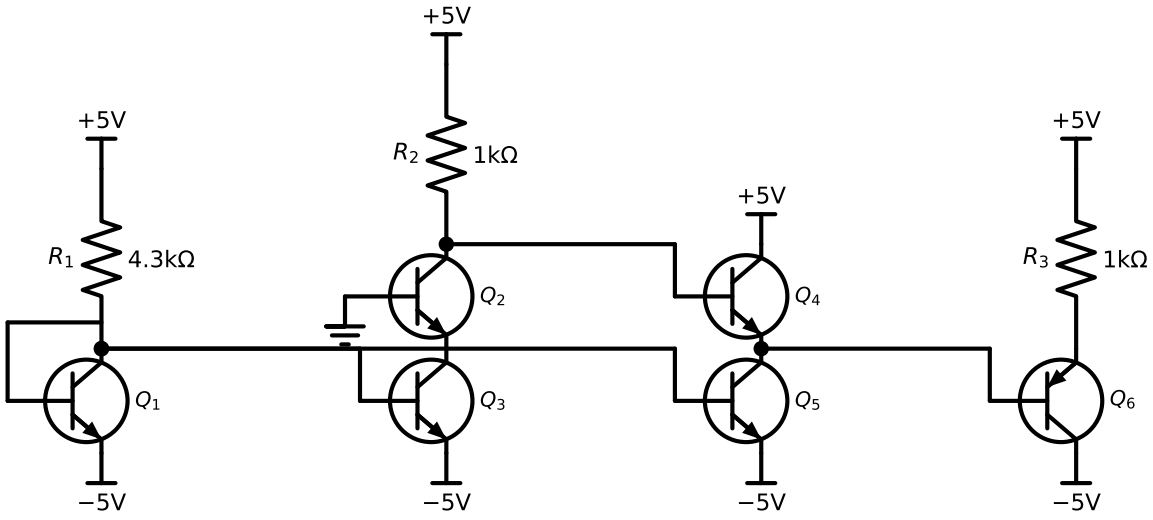
Problem 4B.17 — Hard

For the circuit below, find the DC collector currents I_C of all transistors and verify all assumptions, including the operating region of every transistor. Assume $\beta = 100$.



Problem 4B.18 — Hard

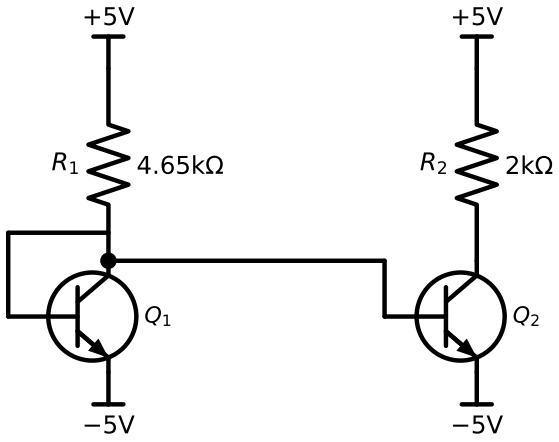
For the circuit below, find the DC collector currents I_C of all transistors and verify all assumptions, including the operating region of every transistor. Assume $\beta = 100$.



Solutions

Every solution follows the steps of the Lecture-4 pipeline card.

Solution 4B.1 — Easy



Step 1 — Assumptions

Assume every transistor is in the active region and that every base current is negligible ($I_B \approx 0$, tool T1): then $|V_{BE}| = 0.7\,\text{V}$, $I_C \approx I_E$, and no current flows into any base, so a base network behaves like a plain resistor divider.

Step 2 — Walk through the circuit

1. Q_1 (diode-connected) is active: $V_{B1} = (-5) + 0.7 = -4.3\,\text{V}$.
2. KCL at the collector of Q_1 (base current neglected):

$$I_{C1} = \frac{5 - V_{B1}}{R_1} = 2\,\text{mA}$$

3. Q_2 has the same base and emitter nodes as Q_1, hence the same V_{BE} and the same current (current mirror): $I_{C2} = I_{C1} = 2\,\text{mA}$.
4. Q_2 draws $I_{C2} = 2\,\text{mA}$ from V_{C2} through R_2: $V_{C2} = 5 - R_2\,(I_{C2}) = 5 - 2\,\text{k} \times 2\,\text{mA} = 1\,\text{V}$.

Step 3 — Verify the assumptions

- $I_{B1} = \frac{I_{C1}}{\beta} = 20\,\mu\text{A}$ (all bases on this node: $40\,\mu\text{A}$) $\ll$ the $2\,\text{mA}$ flowing in the base network ✓
- $I_{B2} = \frac{I_{C2}}{\beta} = 20\,\mu\text{A}$ (all bases on this node: $40\,\mu\text{A}$) $\ll$ the $2\,\text{mA}$ flowing in the base network ✓
- $V_{CE1} = V_{B1} - (-5) = -4.3 - (-5) = 0.7\,\text{V} \geq 0.2\,\text{V} \Rightarrow Q_1$ is active ✓
- $V_{CE2} = V_{C2} - (-5) = 1 - (-5) = 6\,\text{V} \geq 0.2\,\text{V} \Rightarrow Q_2$ is active ✓

Step 4 — How good was $I_B \approx 0$?

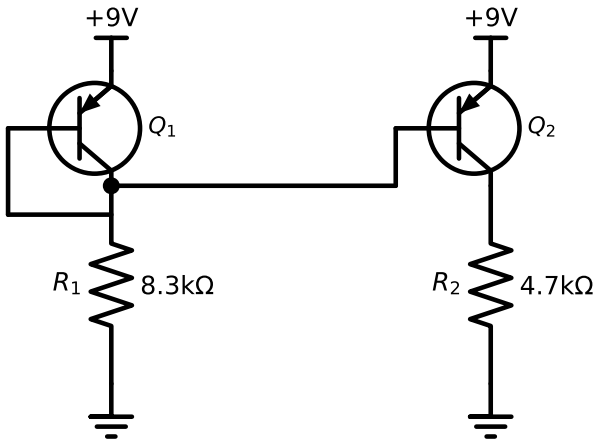
If we keep the base currents ($I_C = \beta I_B$ with $\beta = 100$) and solve the circuit exactly, the currents change only a little:

transistor	I_C (with $I_B \approx 0$)	I_C (exact)	difference
Q_1	2 mA	1.96 mA	2%
Q_2	2 mA	1.96 mA	2%

Answer

$I_{C1} = 2\,\text{mA}, \quad I_{C2} = 2\,\text{mA}$
all transistors active, all base currents negligible.

Solution 4B.2 **Easy**



Step 1 — Assumptions

Assume every transistor is in the active region and that every base current is negligible ($I_B \approx 0$, tool T1): then $|V_{BE}| = 0.7\,\text{V}$, $I_C \approx I_E$, and no current flows into any base, so a base network behaves like a plain resistor divider.

Step 2 — Walk through the circuit

1. Q_1 (diode-connected) is active: $V_{B1} = 9 - 0.7 = 8.3\,\text{V}$.
2. KCL at the collector of Q_1 (base current neglected):
$$I_{C1} = \frac{V_{B1} - 0}{R_1} = 1\,\text{mA}$$
3. Q_2 has the same base and emitter nodes as Q_1, hence the same V_{BE} and the same current (current mirror): $I_{C2} = I_{C1} = 1\,\text{mA}$.
4. The current $I_{C2} = 1\,\text{mA}$ flows from V_{C2} through R_2: $V_{C2} = 0 + R_2\,(I_{C2}) = 0 + 4.7\,\text{k} \times 1\,\text{mA} = 4.7\,\text{V}$.

Step 3 — Verify the assumptions

- $I_{B1} = \frac{I_{C1}}{\beta} = 10\,\mu\text{A}$ (all bases on this node: $20\,\mu\text{A}$) $\ll$ the $1\,\text{mA}$ flowing in the base network ✓
- $I_{B2} = \frac{I_{C2}}{\beta} = 10\,\mu\text{A}$ (all bases on this node: $20\,\mu\text{A}$) $\ll$ the $1\,\text{mA}$ flowing in the base network ✓
- $V_{EC1} = 9 - V_{B1} = 9 - (8.3) = 0.7\,\text{V} \geq 0.2\,\text{V} \Rightarrow Q_1$ is active ✓
- $V_{EC2} = 9 - V_{C2} = 9 - (4.7) = 4.3\,\text{V} \geq 0.2\,\text{V} \Rightarrow Q_2$ is active ✓

Step 4 — How good was $I_B \approx 0$?

If we keep the base currents ($I_C = \beta I_B$ with $\beta = 100$) and solve the circuit exactly, the currents change only a little:

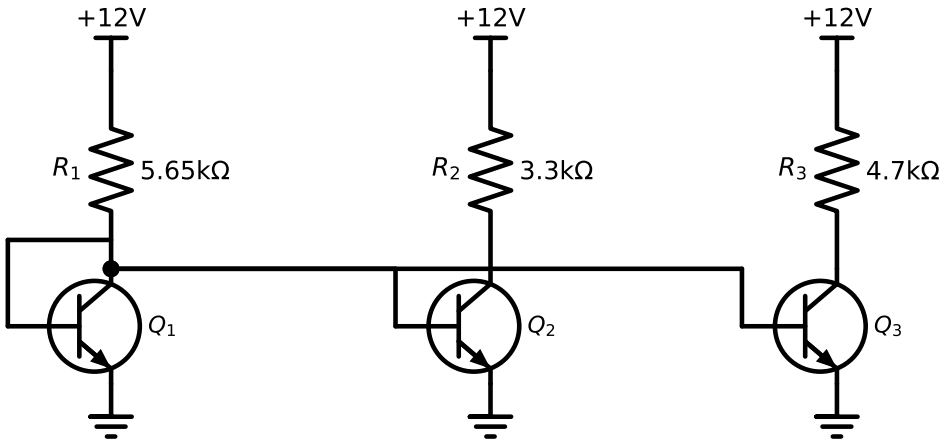
transistor	I_C (with $I_B \approx 0$)	I_C (exact)	difference
Q_1	$1\,\text{mA}$	$980\,\mu\text{A}$	2%
Q_2	$1\,\text{mA}$	$980\,\mu\text{A}$	2%

Answer

$I_{C1} = 1\,\text{mA}, \quad I_{C2} = 1\,\text{mA}$

all transistors active, all base currents negligible.

Solution 4B.3 **Easy**



Step 1 — Assumptions

Assume every transistor is in the active region and that every base current is negligible ($I_B \approx 0$, tool T1): then $|V_{BE}| = 0.7\,\text{V}$, $I_C \approx I_E$, and no current flows into any base, so a base network behaves like a plain resistor divider.

Step 2 — Walk through the circuit

1. Q_1 (diode-connected) is active: $V_{B1} = 0 + 0.7 = 0.7\,\text{V}$.
2. KCL at the collector of Q_1 (base current neglected):

$$I_{C1} = \frac{12 - V_{B1}}{R_1} = 2\,\text{mA}$$

3. Q_2 has the same base and emitter nodes as Q_1, hence the same V_{BE} and the same current (current mirror): $I_{C2} = I_{C1} = 2\,\text{mA}$.
4. Q_3 has the same base and emitter nodes as Q_1, hence the same V_{BE} and the same current (current mirror): $I_{C3} = I_{C1} = 2\,\text{mA}$.
5. Q_2 draws $I_{C2} = 2\,\text{mA}$ from V_{C2} through R_2: $V_{C2} = 12 - R_2\,(I_{C2}) = 12 - 3.3\,\text{k} \times 2\,\text{mA} = 5.4\,\text{V}$.
6. Q_3 draws $I_{C3} = 2\,\text{mA}$ from V_{C3} through R_3: $V_{C3} = 12 - R_3\,(I_{C3}) = 12 - 4.7\,\text{k} \times 2\,\text{mA} = 2.6\,\text{V}$.

Step 3 — Verify the assumptions

- $I_{B1} = \frac{I_{C1}}{\beta} = 20\,\mu\text{A}$ (all bases on this node: $60\,\mu\text{A}$) $\ll$ the $2\,\text{mA}$ flowing in the base network ✓
- $I_{B2} = \frac{I_{C2}}{\beta} = 20\,\mu\text{A}$ (all bases on this node: $60\,\mu\text{A}$) $\ll$ the $2\,\text{mA}$ flowing in the base network ✓
- $I_{B3} = \frac{I_{C3}}{\beta} = 20\,\mu\text{A}$ (all bases on this node: $60\,\mu\text{A}$) $\ll$ the $2\,\text{mA}$ flowing in the base network ✓
- $V_{CE1} = V_{B1} - 0 = 0.7 - (0) = 0.7\,\text{V} \geq 0.2\,\text{V} \Rightarrow Q_1$ is active ✓
- $V_{CE2} = V_{C2} - 0 = 5.4 - (0) = 5.4\,\text{V} \geq 0.2\,\text{V} \Rightarrow Q_2$ is active ✓
- $V_{CE3} = V_{C3} - 0 = 2.6 - (0) = 2.6\,\text{V} \geq 0.2\,\text{V} \Rightarrow Q_3$ is active ✓

Step 4 — How good was $I_B \approx 0$?

If we keep the base currents ($I_C = \beta I_B$ with $\beta = 100$) and solve the circuit exactly, the currents change only a little:

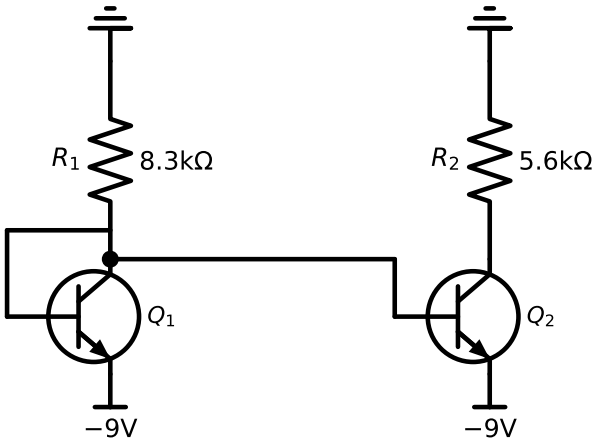
transistor	I_C (with $I_B \approx 0$)	I_C (exact)	difference
Q_1	2 mA	1.94 mA	3%
Q_2	2 mA	1.94 mA	3%
Q_3	2 mA	1.94 mA	3%

Answer

$I_{C1} = 2\,\text{mA}, \quad I_{C2} = 2\,\text{mA}, \quad I_{C3} = 2\,\text{mA}$

all transistors active, all base currents negligible.

Solution 4B.4 **Easy**



Step 1 — Assumptions

Assume every transistor is in the active region and that every base current is negligible ($I_B \approx 0$, tool T1): then $|V_{BE}| = 0.7\,\text{V}$, $I_C \approx I_E$, and no current flows into any base, so a base network behaves like a plain resistor divider.

Step 2 — Walk through the circuit

1. Q_1 (diode-connected) is active: $V_{B1} = (-9) + 0.7 = -8.3\,\text{V}$.
2. KCL at the collector of Q_1 (base current neglected):
$$I_{C1} = \frac{0 - V_{B1}}{R_1} = 1\,\text{mA}$$
3. Q_2 has the same base and emitter nodes as Q_1, hence the same V_{BE} and the same current (current mirror): $I_{C2} = I_{C1} = 1\,\text{mA}$.
4. Q_2 draws $I_{C2} = 1\,\text{mA}$ from V_{C2} through R_2: $V_{C2} = 0 - R_2\,(I_{C2}) = 0 - 5.6\,\text{k} \times 1\,\text{mA} = -5.6\,\text{V}$.

Step 3 — Verify the assumptions

- $I_{B1} = \frac{I_{C1}}{\beta} = 10\,\mu\text{A}$ (all bases on this node: $20\,\mu\text{A}$) $\ll$ the $1\,\text{mA}$ flowing in the base network ✓
- $I_{B2} = \frac{I_{C2}}{\beta} = 10\,\mu\text{A}$ (all bases on this node: $20\,\mu\text{A}$) $\ll$ the $1\,\text{mA}$ flowing in the base network ✓
- $V_{CE1} = V_{B1} - (-9) = -8.3 - (-9) = 0.7\,\text{V} \geq 0.2\,\text{V} \Rightarrow Q_1$ is active ✓
- $V_{CE2} = V_{C2} - (-9) = -5.6 - (-9) = 3.4\,\text{V} \geq 0.2\,\text{V} \Rightarrow Q_2$ is active ✓

Step 4 — How good was $I_B \approx 0$?

If we keep the base currents ($I_C = \beta I_B$ with $\beta = 100$) and solve the circuit exactly, the currents change only a little:

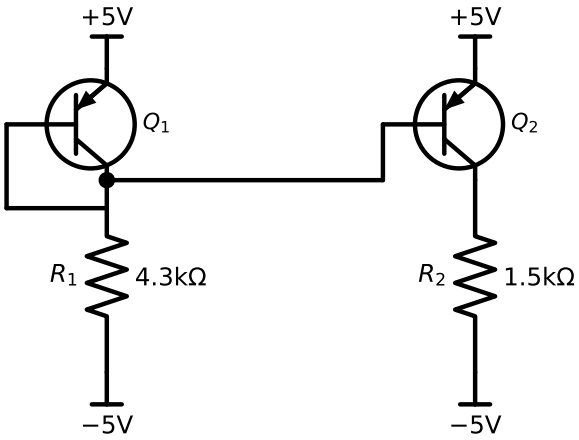
transistor	I_C (with $I_B \approx 0$)	I_C (exact)	difference
Q_1	$1\,\text{mA}$	$980\,\mu\text{A}$	2%
Q_2	$1\,\text{mA}$	$980\,\mu\text{A}$	2%

Answer

$I_{C1} = 1\,\text{mA}, \quad I_{C2} = 1\,\text{mA}$

all transistors active, all base currents negligible.

Solution 4B.5 **Easy**



Step 1 — Assumptions

Assume every transistor is in the active region and that every base current is negligible ($I_B \approx 0$, tool T1): then $|V_{BE}| = 0.7\,\text{V}$, $I_C \approx I_E$, and no current flows into any base, so a base network behaves like a plain resistor divider.

Step 2 — Walk through the circuit

1. Q_1 (diode-connected) is active: $V_{B1} = 5 - 0.7 = 4.3\,\text{V}$.
2. KCL at the collector of Q_1 (base current neglected):

$$I_{C1} = \frac{V_{B1} - (-5)}{R_1} = 2.16\,\text{mA}$$

3. Q_2 has the same base and emitter nodes as Q_1, hence the same V_{BE} and the same current (current mirror): $I_{C2} = I_{C1} = 2.16\,\text{mA}$.
4. The current $I_{C2} = 2.16\,\text{mA}$ flows from V_{C2} through R_2: $V_{C2} = (-5) + R_2\,(I_{C2}) = -5 + 1.5\,\text{k} \times 2.16\,\text{mA} = -1.76\,\text{V}$.

Step 3 — Verify the assumptions

- $I_{B1} = \frac{I_{C1}}{\beta} = 21.6\,\mu\text{A}$ (all bases on this node: $43.3\,\mu\text{A}$) $\ll$ the $2.16\,\text{mA}$ flowing in the base network ✓
- $I_{B2} = \frac{I_{C2}}{\beta} = 21.6\,\mu\text{A}$ (all bases on this node: $43.3\,\mu\text{A}$) $\ll$ the $2.16\,\text{mA}$ flowing in the base network ✓
- $V_{EC1} = 5 - V_{B1} = 5 - (4.3) = 0.7\,\text{V} \geq 0.2\,\text{V} \Rightarrow Q_1$ is active ✓
- $V_{EC2} = 5 - V_{C2} = 5 - (-1.756) = 6.76\,\text{V} \geq 0.2\,\text{V} \Rightarrow Q_2$ is active ✓

Step 4 — How good was $I_B \approx 0$?

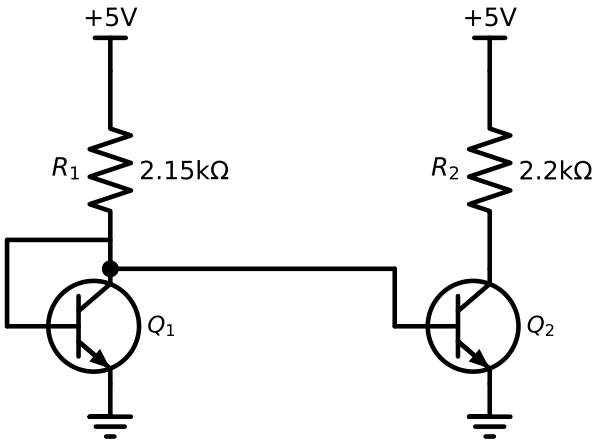
If we keep the base currents ($I_C = \beta I_B$ with $\beta = 100$) and solve the circuit exactly, the currents change only a little:

transistor	I_C (with $I_B \approx 0$)	I_C (exact)	difference
Q_1	2.16 mA	2.12 mA	2%
Q_2	2.16 mA	2.12 mA	2%

Answer

$I_{C1} = 2.16\,\text{mA}, \quad I_{C2} = 2.16\,\text{mA}$
all transistors active, all base currents negligible.

Solution 4B.6 **Easy**



Step 1 — Assumptions

Assume every transistor is in the active region and that every base current is negligible ($I_B \approx 0$, tool T1): then $|V_{BE}| = 0.7\,\text{V}$, $I_C \approx I_E$, and no current flows into any base, so a base network behaves like a plain resistor divider.

Step 2 — Walk through the circuit

1. Q_1 (diode-connected) is active: $V_{B1} = 0 + 0.7 = 0.7\,\text{V}$.
2. KCL at the collector of Q_1 (base current neglected):
$$I_{C1} = \frac{5 - V_{B1}}{R_1} = 2\,\text{mA}$$
3. Q_2 has the same base and emitter nodes as Q_1, hence the same V_{BE} and the same current (current mirror): $I_{C2} = I_{C1} = 2\,\text{mA}$.
4. Q_2 draws $I_{C2} = 2\,\text{mA}$ from V_{C2} through R_2: $V_{C2} = 5 - R_2\,(I_{C2}) = 5 - 2.2\,\text{k} \times 2\,\text{mA} = 0.6\,\text{V}$.

Step 3 — Verify the assumptions

- $I_{B1} = \frac{I_{C1}}{\beta} = 20\,\mu\text{A}$ (all bases on this node: $40\,\mu\text{A}$) $\ll$ the $2\,\text{mA}$ flowing in the base network ✓
- $I_{B2} = \frac{I_{C2}}{\beta} = 20\,\mu\text{A}$ (all bases on this node: $40\,\mu\text{A}$) $\ll$ the $2\,\text{mA}$ flowing in the base network ✓
- $V_{CE1} = V_{B1} - 0 = 0.7 - (0) = 0.7\,\text{V} \geq 0.2\,\text{V} \Rightarrow Q_1$ is active ✓
- $V_{CE2} = V_{C2} - 0 = 0.6 - (0) = 0.6\,\text{V} \geq 0.2\,\text{V} \Rightarrow Q_2$ is active ✓

Step 4 — How good was $I_B \approx 0$?

If we keep the base currents ($I_C = \beta I_B$ with $\beta = 100$) and solve the circuit exactly, the currents change only a little:

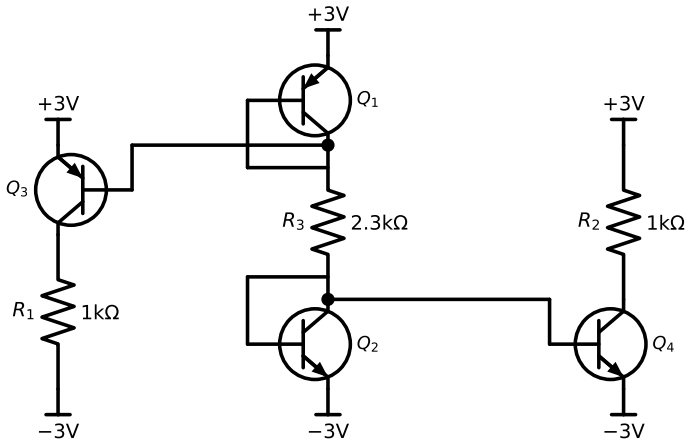
transistor	I_C (with $I_B \approx 0$)	I_C (exact)	difference
Q_1	$2\,\text{mA}$	$1.96\,\text{mA}$	2%
Q_2	$2\,\text{mA}$	$1.96\,\text{mA}$	2%

Answer

$I_{C1} = 2\,\text{mA}, \quad I_{C2} = 2\,\text{mA}$

all transistors active, all base currents negligible.

Solution 4B.7 **Medium**



Step 1 — Assumptions

Assume every transistor is in the active region and that every base current is negligible ($I_B \approx 0$, tool T1): then $|V_{BE}| = 0.7\,\text{V}$, $I_C \approx I_E$, and no current flows into any base, so a base network behaves like a plain resistor divider.

Step 2 — Walk through the circuit

1. Q_3 is active: $V_{B3} = 3 - 0.7 = 2.3\,\text{V}$.
2. Q_2 (diode-connected) is active: $V_{B2} = (-3) + 0.7 = -2.3\,\text{V}$.
3. KCL at the collector of Q_2 (base current neglected):

$$I_{C2} = \frac{V_{B3} - V_{B2}}{R_3} = 2\,\text{mA}$$

4. KCL at the collector of Q_1 (base current neglected):

$$I_{C1} = \frac{V_{B3} - V_{B2}}{R_3} = 2\,\text{mA}$$

5. Q_3 has the same base and emitter nodes as Q_1, hence the same V_{BE} and the same current (current mirror): $I_{C3} = I_{C1} = 2\,\text{mA}$.
6. Q_4 has the same base and emitter nodes as Q_2, hence the same V_{BE} and the same current (current mirror): $I_{C4} = I_{C2} = 2\,\text{mA}$.
7. The current $I_{C3} = 2\,\text{mA}$ flows from V_{C3} through R_1: $V_{C3} = (-3) + R_1\,(I_{C3}) = -3 + 1\,\text{k} \times 2\,\text{mA} = -1\,\text{V}$.
8. Q_4 draws $I_{C4} = 2\,\text{mA}$ from V_{C4} through R_2: $V_{C4} = 3 - R_2\,(I_{C4}) = 3 - 1\,\text{k} \times 2\,\text{mA} = 1\,\text{V}$.

Step 3 — Verify the assumptions

- $I_{B3} = \frac{I_{C3}}{\beta} = 20\,\mu\text{A}$ (all bases on this node: $40\,\mu\text{A}$) $\ll$ the $2\,\text{mA}$ flowing in the base network ✓
- $I_{B1} = \frac{I_{C1}}{\beta} = 20\,\mu\text{A}$ (all bases on this node: $40\,\mu\text{A}$) $\ll$ the $2\,\text{mA}$ flowing in the base network ✓

- $I_{B2} = \frac{I_{C2}}{\beta} = 20\,\mu\text{A}$ (all bases on this node: $40\,\mu\text{A}$) $\ll$ the 2 mA flowing in the base network ✓
- $I_{B4} = \frac{I_{C4}}{\beta} = 20\,\mu\text{A}$ (all bases on this node: $40\,\mu\text{A}$) $\ll$ the 2 mA flowing in the base network ✓
- $V_{EC3} = 3 - V_{C3} = 3 - (-1) = 4\,\text{V} \geq 0.2\,\text{V} \Rightarrow Q_3$ is active ✓
- $V_{EC1} = 3 - V_{B3} = 3 - (2.3) = 0.7\,\text{V} \geq 0.2\,\text{V} \Rightarrow Q_1$ is active ✓
- $V_{CE2} = V_{B2} - (-3) = -2.3 - (-3) = 0.7\,\text{V} \geq 0.2\,\text{V} \Rightarrow Q_2$ is active ✓
- $V_{CE4} = V_{C4} - (-3) = 1 - (-3) = 4\,\text{V} \geq 0.2\,\text{V} \Rightarrow Q_4$ is active ✓

Step 4 — How good was $I_B \approx 0$?

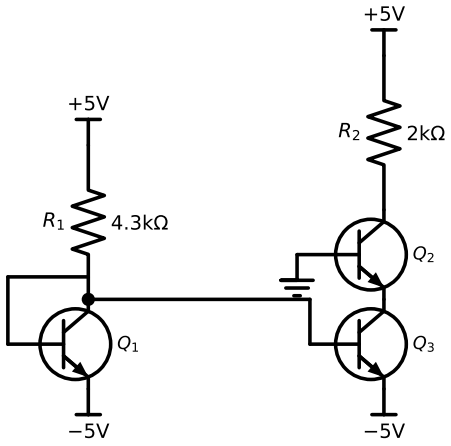
If we keep the base currents ($I_C = \beta I_B$ with $\beta = 100$) and solve the circuit exactly, the currents change only a little:

transistor	I_C (with $I_B \approx 0$)	I_C (exact)	difference
Q_3	2 mA	1.96 mA	2%
Q_1	2 mA	1.96 mA	2%
Q_2	2 mA	1.96 mA	2%
Q_4	2 mA	1.96 mA	2%

Answer

$I_{C3} = 2\,\text{mA}, \quad I_{C1} = 2\,\text{mA}, \quad I_{C2} = 2\,\text{mA}, \quad I_{C4} = 2\,\text{mA}$
all transistors active, all base currents negligible.

Solution 4B.8 **Medium**



Step 1 — Assumptions

Assume every transistor is in the active region and that every base current is negligible ($I_B \approx 0$, tool T1): then $|V_{BE}| = 0.7\,\text{V}$, $I_C \approx I_E$, and no current flows into any base, so a base network behaves like a plain resistor divider.

Step 2 — Walk through the circuit

1. Q_1 (diode-connected) is active: $V_{B1} = (-5) + 0.7 = -4.3\,\text{V}$.
2. Q_2 is npn and assumed active: $V_{BE} = 0.7\,\text{V}$, so $V_{E2} = 0 - 0.7 = -0.7\,\text{V}$.
3. KCL at the collector of Q_1 (base current neglected):

$$I_{C1} = \frac{5 - V_{B1}}{R_1} = 2.16\,\text{mA}$$

4. Q_3 has the same base and emitter nodes as Q_1, hence the same V_{BE} and the same current (current mirror): $I_{C3} = I_{C1} = 2.16\,\text{mA}$.
5. KCL at the emitter of Q_2 (base current neglected):

$$I_{E2} = I_{C3} = 2.16\,\text{mA}, \qquad I_{C2} \approx I_{E2}$$

6. Q_2 draws $I_{C2} = 2.16\,\text{mA}$ from V_{C2} through R_2: $V_{C2} = 5 - R_2\,(I_{C2}) = 5 - 2\,\text{k} \times 2.16\,\text{mA} = 0.674\,\text{V}$.

Step 3 — Verify the assumptions

- $I_{B1} = \frac{I_{C1}}{\beta} = 21.6\,\mu\text{A}$ (all bases on this node: $43.3\,\mu\text{A}$) $\ll$ the $2.16\,\text{mA}$ flowing in the base network ✓
- $I_{B3} = \frac{I_{C3}}{\beta} = 21.6\,\mu\text{A}$ (all bases on this node: $43.3\,\mu\text{A}$) $\ll$ the $2.16\,\text{mA}$ flowing in the base network ✓
- $V_{CE1} = V_{B1} - (-5) = -4.3 - (-5) = 0.7\,\text{V} \geq 0.2\,\text{V} \Rightarrow Q_1$ is active ✓
- $V_{CE2} = V_{C2} - V_{E2} = 0.6744 - (-0.7) = 1.37\,\text{V} \geq 0.2\,\text{V} \Rightarrow Q_2$ is active ✓
- $V_{CE3} = V_{E2} - (-5) = -0.7 - (-5) = 4.3\,\text{V} \geq 0.2\,\text{V} \Rightarrow Q_3$ is active ✓

Step 4 — How good was $I_B \approx 0$?

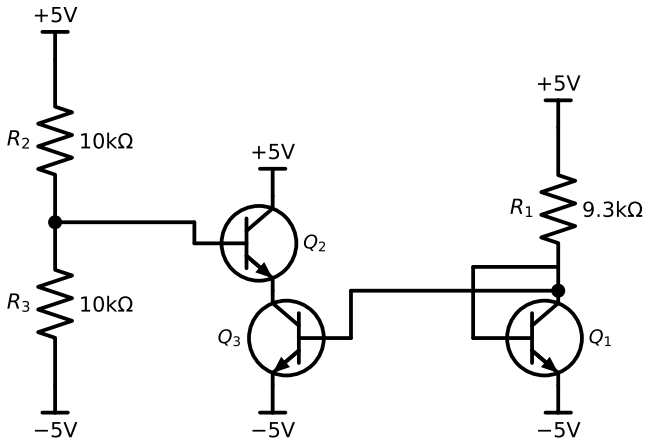
If we keep the base currents ($I_C = \beta I_B$ with $\beta = 100$) and solve the circuit exactly, the currents change only a little:

transistor	I_C (with $I_B \approx 0$)	I_C (exact)	difference
Q_1	2.16 mA	2.12 mA	2%
Q_2	2.16 mA	2.1 mA	3%
Q_3	2.16 mA	2.12 mA	2%

Answer

$I_{C1} = 2.16\,\text{mA}, \quad I_{C2} = 2.16\,\text{mA}, \quad I_{C3} = 2.16\,\text{mA}$
all transistors active, all base currents negligible.

Solution 4B.9 **Medium**



Step 1 — Assumptions

Assume every transistor is in the active region and that every base current is negligible ($I_B \approx 0$, tool T1): then $|V_{BE}| = 0.7\,\text{V}$, $I_C \approx I_E$, and no current flows into any base, so a base network behaves like a plain resistor divider.

Step 2 — Walk through the circuit

1. Q_3 is active: $V_{B3} = (-5) + 0.7 = -4.3\,\text{V}$.
2. No current flows into the base (I_B is negligible), so V_{B2} is set by the divider R_2–R_3 between 5 and (-5) (Lecture-1 tool):
$$V_{B2} = -5 + (5 - (-5))\frac{10\,\text{k}}{10\,\text{k} + 10\,\text{k}} = 0\,\text{V}$$
3. KCL at the collector of Q_1 (base current neglected):
$$I_{C1} = \frac{5 - V_{B3}}{R_1} = 1\,\text{mA}$$
4. Q_2 is npn and assumed active: $V_{BE} = 0.7\,\text{V}$, so $V_{E2} = V_{B2} - 0.7 = -0.7\,\text{V}$.
5. Q_3 has the same base and emitter nodes as Q_1, hence the same V_{BE} and the same current (current mirror): $I_{C3} = I_{C1} = 1\,\text{mA}$.
6. KCL at the emitter of Q_2 (base current neglected):
$$I_{E2} = I_{C3} = 1\,\text{mA}, \qquad I_{C2} \approx I_{E2}$$

Step 3 — Verify the assumptions

- $I_{B2} = \frac{I_{C2}}{\beta} = 10\,\mu\text{A} \ll$ the $500\,\mu\text{A}$ flowing in the base network ✓
- $I_{B3} = \frac{I_{C3}}{\beta} = 10\,\mu\text{A}$ (all bases on this node: $20\,\mu\text{A}$) $\ll$ the 1 mA flowing in the base network ✓
- $I_{B1} = \frac{I_{C1}}{\beta} = 10\,\mu\text{A}$ (all bases on this node: $20\,\mu\text{A}$) $\ll$ the 1 mA flowing in the base network ✓
- $V_{CE2} = 5 - V_{E2} = 5 - (-0.7) = 5.7\,\text{V} \geq 0.2\,\text{V} \Rightarrow Q_2$ is active ✓

- $V_{CE3} = V_{E2} - (-5) = -0.7 - (-5) = 4.3\,\text{V} \geq 0.2\,\text{V} \Rightarrow Q_3$ is active ✓
- $V_{CE1} = V_{B3} - (-5) = -4.3 - (-5) = 0.7\,\text{V} \geq 0.2\,\text{V} \Rightarrow Q_1$ is active ✓

Step 4 — How good was $I_B \approx 0$?

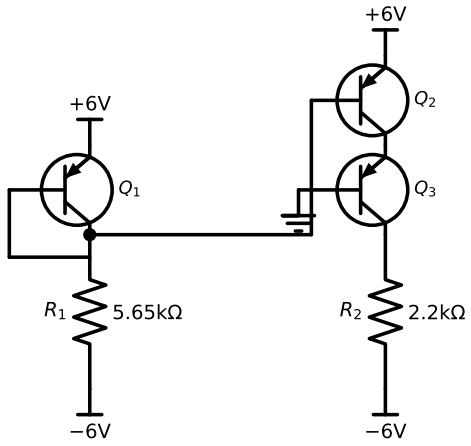
If we keep the base currents ($I_C = \beta I_B$ with $\beta = 100$) and solve the circuit exactly, the currents change only a little:

transistor	I_C (with $I_B \approx 0$)	I_C (exact)	difference
Q_2	$1\,\text{mA}$	$971\,\mu\text{A}$	3%
Q_3	$1\,\text{mA}$	$980\,\mu\text{A}$	2%
Q_1	$1\,\text{mA}$	$980\,\mu\text{A}$	2%

Answer

$I_{C2} = 1\,\text{mA}, \quad I_{C3} = 1\,\text{mA}, \quad I_{C1} = 1\,\text{mA}$
all transistors active, all base currents negligible.

Solution 4B.10 **Medium**



Step 1 — Assumptions

Assume every transistor is in the active region and that every base current is negligible ($I_B \approx 0$, tool T1): then $|V_{BE}| = 0.7\,\text{V}$, $I_C \approx I_E$, and no current flows into any base, so a base network behaves like a plain resistor divider.

Step 2 — Walk through the circuit

1. Q_1 (diode-connected) is active: $V_{B1} = 6 - 0.7 = 5.3\,\text{V}$.
2. Q_3 is pnp and assumed active: $V_{BE} = -0.7\,\text{V}$, so $V_{C2} = 0 + 0.7 = 0.7\,\text{V}$.
3. KCL at the collector of Q_1 (base current neglected):

$$I_{C1} = \frac{V_{B1} - (-6)}{R_1} = 2\,\text{mA}$$

4. Q_2 has the same base and emitter nodes as Q_1, hence the same V_{BE} and the same current (current mirror): $I_{C2} = I_{C1} = 2\,\text{mA}$.
5. KCL at the emitter of Q_3 (base current neglected):

$$I_{E3} = I_{C2} = 2\,\text{mA}, \qquad I_{C3} \approx I_{E3}$$

6. The current $I_{C3} = 2\,\text{mA}$ flows from V_{C3} through R_2: $V_{C3} = (-6) + R_2\,(I_{C3}) = -6 + 2.2\,\text{k} \times 2\,\text{mA} = -1.6\,\text{V}$.

Step 3 — Verify the assumptions

- $I_{B1} = \frac{I_{C1}}{\beta} = 20\,\mu\text{A}$ (all bases on this node: $40\,\mu\text{A}$) $\ll$ the $2\,\text{mA}$ flowing in the base network ✓
- $I_{B2} = \frac{I_{C2}}{\beta} = 20\,\mu\text{A}$ (all bases on this node: $40\,\mu\text{A}$) $\ll$ the $2\,\text{mA}$ flowing in the base network ✓
- $V_{EC1} = 6 - V_{B1} = 6 - (5.3) = 0.7\,\text{V} \geq 0.2\,\text{V} \Rightarrow Q_1$ is active ✓
- $V_{EC2} = 6 - V_{C2} = 6 - (0.7) = 5.3\,\text{V} \geq 0.2\,\text{V} \Rightarrow Q_2$ is active ✓
- $V_{EC3} = V_{C2} - V_{C3} = 0.7 - (-1.6) = 2.3\,\text{V} \geq 0.2\,\text{V} \Rightarrow Q_3$ is active ✓

Step 4 — How good was $I_B \approx 0$?

If we keep the base currents ($I_C = \beta I_B$ with $\beta = 100$) and solve the circuit exactly, the currents change only a little:

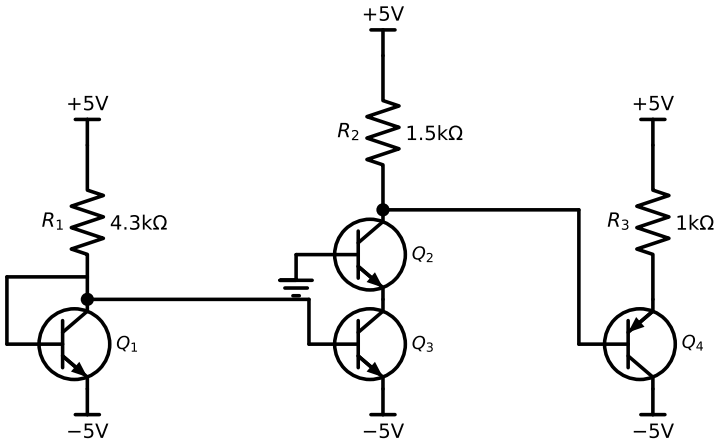
transistor	I_C (with $I_B \approx 0$)	I_C (exact)	difference
Q_1	2 mA	1.96 mA	2%
Q_2	2 mA	1.96 mA	2%
Q_3	2 mA	1.94 mA	3%

Answer

$I_{C1} = 2\,\text{mA}, \quad I_{C2} = 2\,\text{mA}, \quad I_{C3} = 2\,\text{mA}$

all transistors active, all base currents negligible.

Solution 4B.11 **Medium**



Step 1 — Assumptions

Assume every transistor is in the active region and that every base current is negligible ($I_B \approx 0$, tool T1): then $|V_{BE}| = 0.7\,\text{V}$, $I_C \approx I_E$, and no current flows into any base, so a base network behaves like a plain resistor divider.

Step 2 — Walk through the circuit

1. Q_1 (diode-connected) is active: $V_{B1} = (-5) + 0.7 = -4.3\,\text{V}$.
2. Q_2 is npn and assumed active: $V_{BE} = 0.7\,\text{V}$, so $V_{E2} = 0 - 0.7 = -0.7\,\text{V}$.
3. KCL at the collector of Q_1 (base current neglected):

$$I_{C1} = \frac{5 - V_{B1}}{R_1} = 2.16\,\text{mA}$$

4. Q_3 has the same base and emitter nodes as Q_1, hence the same V_{BE} and the same current (current mirror): $I_{C3} = I_{C1} = 2.16\,\text{mA}$.
5. KCL at the emitter of Q_2 (base current neglected):

$$I_{E2} = I_{C3} = 2.16\,\text{mA}, \qquad I_{C2} \approx I_{E2}$$

6. Q_2 draws $I_{C2} = 2.16\,\text{mA}$ from V_{C2} through R_2: $V_{C2} = 5 - R_2\,(I_{C2}) = 5 - 1.5\,\text{k} \times 2.16\,\text{mA} = 1.76\,\text{V}$.
7. Q_4 is pnp and assumed active: $V_{BE} = -0.7\,\text{V}$, so $V_{E4} = V_{C2} + 0.7 = 2.46\,\text{V}$.
8. KCL at the emitter of Q_4 (base current neglected):

$$I_{E4} = \frac{5 - V_{E4}}{R_3} = 2.54\,\text{mA}, \qquad I_{C4} \approx I_{E4}$$

Step 3 — Verify the assumptions

- $I_{B1} = \frac{I_{C1}}{\beta} = 21.6\,\mu\text{A}$ (all bases on this node: $43.3\,\mu\text{A}$) $\ll$ the $2.16\,\text{mA}$ flowing in the base network ✓
- $I_{B3} = \frac{I_{C3}}{\beta} = 21.6\,\mu\text{A}$ (all bases on this node: $43.3\,\mu\text{A}$) $\ll$ the $2.16\,\text{mA}$ flowing in the base network ✓

- $I_{B4} = \frac{I_{C4}}{\beta} = 25.4\,\mu\text{A} \ll$ the 2.16 mA flowing in the base network ✓
- $V_{CE1} = V_{B1} - (-5) = -4.3 - (-5) = 0.7\,\text{V} \geq 0.2\,\text{V} \Rightarrow Q_1$ is active ✓
- $V_{CE2} = V_{C2} - V_{E2} = 1.756 - (-0.7) = 2.46\,\text{V} \geq 0.2\,\text{V} \Rightarrow Q_2$ is active ✓
- $V_{CE3} = V_{E2} - (-5) = -0.7 - (-5) = 4.3\,\text{V} \geq 0.2\,\text{V} \Rightarrow Q_3$ is active ✓
- $V_{EC4} = V_{E4} - (-5) = 2.456 - (-5) = 7.46\,\text{V} \geq 0.2\,\text{V} \Rightarrow Q_4$ is active ✓

Step 4 — How good was $I_B \approx 0$?

If we keep the base currents ($I_C = \beta I_B$ with $\beta = 100$) and solve the circuit exactly, the currents change only a little:

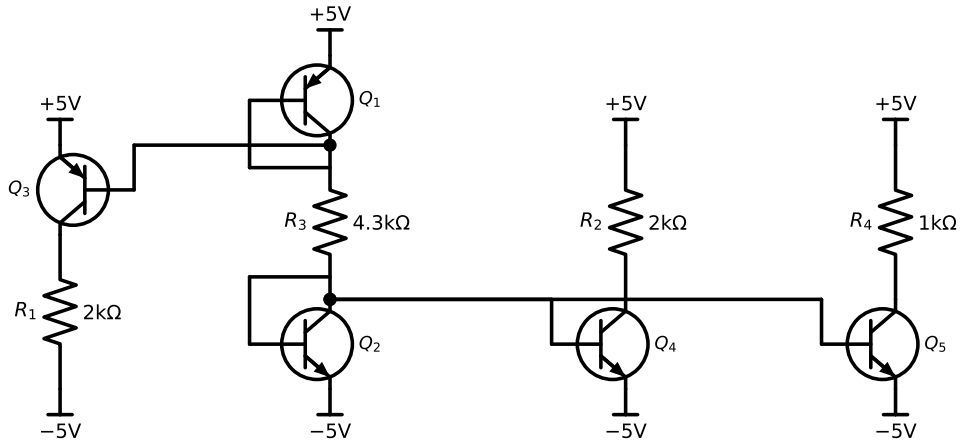
transistor	I_C (with $I_B \approx 0$)	I_C (exact)	difference
Q_1	2.16 mA	2.12 mA	2%
Q_2	2.16 mA	2.1 mA	3%
Q_3	2.16 mA	2.12 mA	2%
Q_4	2.54 mA	2.39 mA	6.5%

The differences grow along the chain: every base current slightly shifts the collector voltage that biases the next transistor, and $\alpha = \beta/(\beta+1)$ makes each I_C about 1 % smaller than its I_E. A shift is amplified when the next stage has only a small voltage across its emitter resistor. Hand analysis with $I_B \approx 0$ is still the standard first pass.

Answer

$I_{C1} = 2.16\,\text{mA}, \quad I_{C2} = 2.16\,\text{mA}, \quad I_{C3} = 2.16\,\text{mA}, \quad I_{C4} = 2.54\,\text{mA}$
all transistors active, all base currents negligible.

Solution 4B.12 **Medium**



Step 1 — Assumptions

Assume every transistor is in the active region and that every base current is negligible ($I_B \approx 0$, tool T1): then $|V_{BE}| = 0.7\,\text{V}$, $I_C \approx I_E$, and no current flows into any base, so a base network behaves like a plain resistor divider.

Step 2 — Walk through the circuit

1. Q_3 is active: $V_{B3} = 5 - 0.7 = 4.3\,\text{V}$.
2. Q_2 (diode-connected) is active: $V_{B2} = (-5) + 0.7 = -4.3\,\text{V}$.
3. KCL at the collector of Q_2 (base current neglected):

$$I_{C2} = \frac{V_{B3} - V_{B2}}{R_3} = 2\,\text{mA}$$

4. KCL at the collector of Q_1 (base current neglected):

$$I_{C1} = \frac{V_{B3} - V_{B2}}{R_3} = 2\,\text{mA}$$

5. Q_3 has the same base and emitter nodes as Q_1, hence the same V_{BE} and the same current (current mirror): $I_{C3} = I_{C1} = 2\,\text{mA}$.
6. Q_4 has the same base and emitter nodes as Q_2, hence the same V_{BE} and the same current (current mirror): $I_{C4} = I_{C2} = 2\,\text{mA}$.
7. Q_5 has the same base and emitter nodes as Q_2, hence the same V_{BE} and the same current (current mirror): $I_{C5} = I_{C2} = 2\,\text{mA}$.
8. The current $I_{C3} = 2\,\text{mA}$ flows from V_{C3} through R_1: $V_{C3} = (-5) + R_1\,(I_{C3}) = -5 + 2\,\text{k} \times 2\,\text{mA} = -1\,\text{V}$.
9. Q_4 draws $I_{C4} = 2\,\text{mA}$ from V_{C4} through R_2: $V_{C4} = 5 - R_2\,(I_{C4}) = 5 - 2\,\text{k} \times 2\,\text{mA} = 1\,\text{V}$.
10. Q_5 draws $I_{C5} = 2\,\text{mA}$ from V_{C5} through R_4: $V_{C5} = 5 - R_4\,(I_{C5}) = 5 - 1\,\text{k} \times 2\,\text{mA} = 3\,\text{V}$.

Step 3 — Verify the assumptions

- $I_{B3} = \frac{I_{C3}}{\beta} = 20\,\mu\text{A}$ (all bases on this node: $40\,\mu\text{A}$) $\ll$ the $2\,\text{mA}$ flowing in the base network ✓
- $I_{B1} = \frac{I_{C1}}{\beta} = 20\,\mu\text{A}$ (all bases on this node: $40\,\mu\text{A}$) $\ll$ the $2\,\text{mA}$ flowing in the base network ✓
- $I_{B2} = \frac{I_{C2}}{\beta} = 20\,\mu\text{A}$ (all bases on this node: $60\,\mu\text{A}$) $\ll$ the $2\,\text{mA}$ flowing in the base network ✓
- $I_{B4} = \frac{I_{C4}}{\beta} = 20\,\mu\text{A}$ (all bases on this node: $60\,\mu\text{A}$) $\ll$ the $2\,\text{mA}$ flowing in the base network ✓
- $I_{B5} = \frac{I_{C5}}{\beta} = 20\,\mu\text{A}$ (all bases on this node: $60\,\mu\text{A}$) $\ll$ the $2\,\text{mA}$ flowing in the base network ✓
- $V_{EC3} = 5 - V_{C3} = 5 - (-1) = 6\,\text{V} \geq 0.2\,\text{V} \Rightarrow Q_3$ is active ✓
- $V_{EC1} = 5 - V_{B3} = 5 - (4.3) = 0.7\,\text{V} \geq 0.2\,\text{V} \Rightarrow Q_1$ is active ✓
- $V_{CE2} = V_{B2} - (-5) = -4.3 - (-5) = 0.7\,\text{V} \geq 0.2\,\text{V} \Rightarrow Q_2$ is active ✓
- $V_{CE4} = V_{C4} - (-5) = 1 - (-5) = 6\,\text{V} \geq 0.2\,\text{V} \Rightarrow Q_4$ is active ✓
- $V_{CE5} = V_{C5} - (-5) = 3 - (-5) = 8\,\text{V} \geq 0.2\,\text{V} \Rightarrow Q_5$ is active ✓

Step 4 — How good was $I_B \approx 0$?

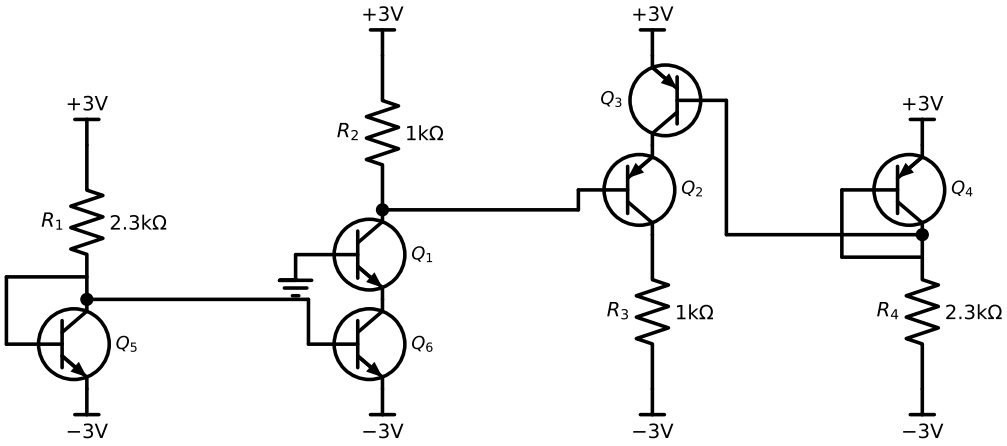
If we keep the base currents ($I_C = \beta I_B$ with $\beta = 100$) and solve the circuit exactly, the currents change only a little:

transistor	I_C (with $I_B \approx 0$)	I_C (exact)	difference
Q_3	$2\,\text{mA}$	$1.96\,\text{mA}$	2%
Q_1	$2\,\text{mA}$	$1.96\,\text{mA}$	2%
Q_2	$2\,\text{mA}$	$1.94\,\text{mA}$	3%
Q_4	$2\,\text{mA}$	$1.94\,\text{mA}$	3%
Q_5	$2\,\text{mA}$	$1.94\,\text{mA}$	3%

Answer

$I_{C3} = 2\,\text{mA}, \quad I_{C1} = 2\,\text{mA}, \quad I_{C2} = 2\,\text{mA}, \quad I_{C4} = 2\,\text{mA}, \quad I_{C5} = 2\,\text{mA}$
all transistors active, all base currents negligible.

Solution 4B.13 **Hard**



Step 1 — Assumptions

Assume every transistor is in the active region and that every base current is negligible ($I_B \approx 0$, tool T1): then $|V_{BE}| = 0.7\,\text{V}$, $I_C \approx I_E$, and no current flows into any base, so a base network behaves like a plain resistor divider.

Step 2 — Walk through the circuit

1. Q_5 (diode-connected) is active: $V_{B5} = (-3) + 0.7 = -2.3\,\text{V}$.
2. Q_1 is npn and assumed active: $V_{BE} = 0.7\,\text{V}$, so $V_{E1} = 0 - 0.7 = -0.7\,\text{V}$.
3. Q_3 is active: $V_{B3} = 3 - 0.7 = 2.3\,\text{V}$.
4. KCL at the collector of Q_5 (base current neglected):

$$I_{C5} = \frac{3 - V_{B5}}{R_1} = 2.3\,\text{mA}$$

5. KCL at the collector of Q_4 (base current neglected):

$$I_{C4} = \frac{V_{B3} - (-3)}{R_4} = 2.3\,\text{mA}$$

6. Q_6 has the same base and emitter nodes as Q_5, hence the same V_{BE} and the same current (current mirror): $I_{C6} = I_{C5} = 2.3\,\text{mA}$.
7. Q_3 has the same base and emitter nodes as Q_4, hence the same V_{BE} and the same current (current mirror): $I_{C3} = I_{C4} = 2.3\,\text{mA}$.
8. KCL at the emitter of Q_1 (base current neglected):

$$I_{E1} = I_{C6} = 2.3\,\text{mA}, \qquad I_{C1} \approx I_{E1}$$

9. Q_1 draws $I_{C1} = 2.3\,\text{mA}$ from V_{C1} through R_2: $V_{C1} = 3 - R_2\,(I_{C1}) = 3 - 1\,\text{k} \times 2.3\,\text{mA} = 0.696\,\text{V}$.
10. Q_2 is pnp and assumed active: $V_{BE} = -0.7\,\text{V}$, so $V_{C3} = V_{C1} + 0.7 = 1.4\,\text{V}$.

11. KCL at the emitter of Q_2 (base current neglected):

$$I_{E2} = I_{C3} = 2.3\,\text{mA}, \qquad I_{C2} \approx I_{E2}$$

12. The current $I_{C2} = 2.3\,\text{mA}$ flows from V_{C2} through R_3: $V_{C2} = (-3) + R_3\,(I_{C2}) = -3 + 1\,\text{k} \times 2.3\,\text{mA} = -0.696\,\text{V}$.

Step 3 — Verify the assumptions

- $I_{B5} = \frac{I_{C5}}{\beta} = 23\,\mu\text{A}$ (all bases on this node: $46.1\,\mu\text{A}$) $\ll$ the $2.3\,\text{mA}$ flowing in the base network ✓
- $I_{B6} = \frac{I_{C6}}{\beta} = 23\,\mu\text{A}$ (all bases on this node: $46.1\,\mu\text{A}$) $\ll$ the $2.3\,\text{mA}$ flowing in the base network ✓
- $I_{B3} = \frac{I_{C3}}{\beta} = 23\,\mu\text{A}$ (all bases on this node: $46.1\,\mu\text{A}$) $\ll$ the $2.3\,\text{mA}$ flowing in the base network ✓
- $I_{B2} = \frac{I_{C2}}{\beta} = 23\,\mu\text{A} \ll$ the $2.3\,\text{mA}$ flowing in the base network ✓
- $I_{B4} = \frac{I_{C4}}{\beta} = 23\,\mu\text{A}$ (all bases on this node: $46.1\,\mu\text{A}$) $\ll$ the $2.3\,\text{mA}$ flowing in the base network ✓
- $V_{CE5} = V_{B5} - (-3) = -2.3 - (-3) = 0.7\,\text{V} \geq 0.2\,\text{V} \Rightarrow Q_5$ is active ✓
- $V_{CE1} = V_{C1} - V_{E1} = 0.6957 - (-0.7) = 1.4\,\text{V} \geq 0.2\,\text{V} \Rightarrow Q_1$ is active ✓
- $V_{CE6} = V_{E1} - (-3) = -0.7 - (-3) = 2.3\,\text{V} \geq 0.2\,\text{V} \Rightarrow Q_6$ is active ✓
- $V_{EC3} = 3 - V_{C3} = 3 - (1.396) = 1.6\,\text{V} \geq 0.2\,\text{V} \Rightarrow Q_3$ is active ✓
- $V_{EC2} = V_{C3} - V_{C2} = 1.396 - (-0.6957) = 2.09\,\text{V} \geq 0.2\,\text{V} \Rightarrow Q_2$ is active ✓
- $V_{EC4} = 3 - V_{B3} = 3 - (2.3) = 0.7\,\text{V} \geq 0.2\,\text{V} \Rightarrow Q_4$ is active ✓

Step 4 — How good was $I_B \approx 0$?

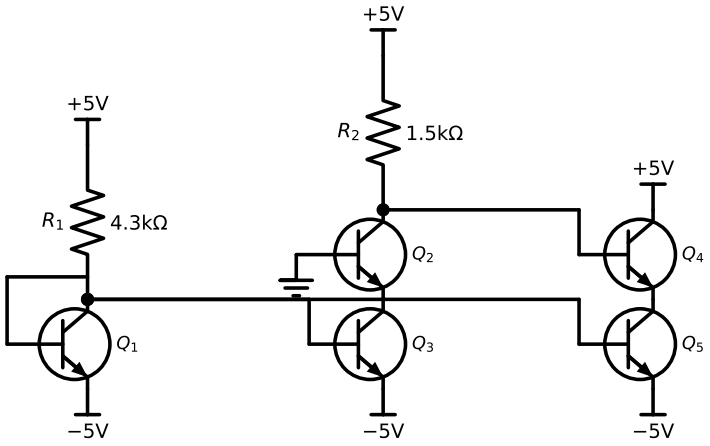
If we keep the base currents ($I_C = \beta I_B$ with $\beta = 100$) and solve the circuit exactly, the currents change only a little:

transistor	I_C (with $I_B \approx 0$)	I_C (exact)	difference
Q_5	$2.3\,\text{mA}$	$2.26\,\text{mA}$	2%
Q_1	$2.3\,\text{mA}$	$2.24\,\text{mA}$	3%
Q_6	$2.3\,\text{mA}$	$2.26\,\text{mA}$	2%
Q_3	$2.3\,\text{mA}$	$2.26\,\text{mA}$	2%
Q_2	$2.3\,\text{mA}$	$2.24\,\text{mA}$	3%
Q_4	$2.3\,\text{mA}$	$2.26\,\text{mA}$	2%

Answer

$I_{C5} = 2.3\,\text{mA}$, $I_{C1} = 2.3\,\text{mA}$, $I_{C6} = 2.3\,\text{mA}$, $I_{C3} = 2.3\,\text{mA}$, $I_{C2} = 2.3\,\text{mA}$, $I_{C4} = 2.3\,\text{mA}$
all transistors active, all base currents negligible.

Solution 4B.14 **Hard**



Step 1 — Assumptions

Assume every transistor is in the active region and that every base current is negligible ($I_B \approx 0$, tool T1): then $|V_{BE}| = 0.7\,\text{V}$, $I_C \approx I_E$, and no current flows into any base, so a base network behaves like a plain resistor divider.

Step 2 — Walk through the circuit

1. Q_1 (diode-connected) is active: $V_{B1} = (-5) + 0.7 = -4.3\,\text{V}$.
2. Q_2 is npn and assumed active: $V_{BE} = 0.7\,\text{V}$, so $V_{E2} = 0 - 0.7 = -0.7\,\text{V}$.
3. KCL at the collector of Q_1 (base current neglected):

$$I_{C1} = \frac{5 - V_{B1}}{R_1} = 2.16\,\text{mA}$$

4. Q_3 has the same base and emitter nodes as Q_1, hence the same V_{BE} and the same current (current mirror): $I_{C3} = I_{C1} = 2.16\,\text{mA}$.
5. Q_5 has the same base and emitter nodes as Q_1, hence the same V_{BE} and the same current (current mirror): $I_{C5} = I_{C1} = 2.16\,\text{mA}$.
6. KCL at the emitter of Q_2 (base current neglected):

$$I_{E2} = I_{C3} = 2.16\,\text{mA}, \qquad I_{C2} \approx I_{E2}$$

7. Q_2 draws $I_{C2} = 2.16\,\text{mA}$ from V_{C2} through R_2: $V_{C2} = 5 - R_2\,(I_{C2}) = 5 - 1.5\,\text{k} \times 2.16\,\text{mA} = 1.76\,\text{V}$.
8. Q_4 is npn and assumed active: $V_{BE} = 0.7\,\text{V}$, so $V_{E4} = V_{C2} - 0.7 = 1.06\,\text{V}$.
9. KCL at the emitter of Q_4 (base current neglected):

$$I_{E4} = I_{C5} = 2.16\,\text{mA}, \qquad I_{C4} \approx I_{E4}$$

Step 3 — Verify the assumptions

- $I_{B1} = \frac{I_{C1}}{\beta} = 21.6\,\mu\text{A}$ (all bases on this node: $64.9\,\mu\text{A}$) $\ll$ the $2.16\,\text{mA}$ flowing in the base network ✓

- $I_{B3} = \frac{I_{C3}}{\beta} = 21.6\,\mu\text{A}$ (all bases on this node: $64.9\,\mu\text{A}$) $\ll$ the 2.16 mA flowing in the base network ✓
- $I_{B4} = \frac{I_{C4}}{\beta} = 21.6\,\mu\text{A} \ll$ the 2.16 mA flowing in the base network ✓
- $I_{B5} = \frac{I_{C5}}{\beta} = 21.6\,\mu\text{A}$ (all bases on this node: $64.9\,\mu\text{A}$) $\ll$ the 2.16 mA flowing in the base network ✓
- $V_{CE1} = V_{B1} - (-5) = -4.3 - (-5) = 0.7\,\text{V} \geq 0.2\,\text{V} \Rightarrow Q_1$ is active ✓
- $V_{CE2} = V_{C2} - V_{E2} = 1.756 - (-0.7) = 2.46\,\text{V} \geq 0.2\,\text{V} \Rightarrow Q_2$ is active ✓
- $V_{CE3} = V_{E2} - (-5) = -0.7 - (-5) = 4.3\,\text{V} \geq 0.2\,\text{V} \Rightarrow Q_3$ is active ✓
- $V_{CE4} = 5 - V_{E4} = 5 - (1.056) = 3.94\,\text{V} \geq 0.2\,\text{V} \Rightarrow Q_4$ is active ✓
- $V_{CE5} = V_{E4} - (-5) = 1.056 - (-5) = 6.06\,\text{V} \geq 0.2\,\text{V} \Rightarrow Q_5$ is active ✓

Step 4 — How good was $I_B \approx 0$?

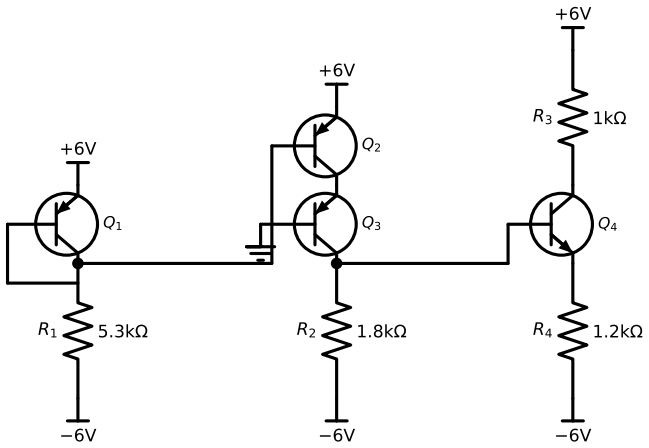
If we keep the base currents ($I_C = \beta I_B$ with $\beta = 100$) and solve the circuit exactly, the currents change only a little:

transistor	I_C (with $I_B \approx 0$)	I_C (exact)	difference
Q_1	2.16 mA	2.1 mA	3%
Q_2	2.16 mA	2.08 mA	4%
Q_3	2.16 mA	2.1 mA	3%
Q_4	2.16 mA	2.08 mA	4%
Q_5	2.16 mA	2.1 mA	3%

Answer

$I_{C1} = 2.16\,\text{mA}$, $I_{C2} = 2.16\,\text{mA}$, $I_{C3} = 2.16\,\text{mA}$, $I_{C4} = 2.16\,\text{mA}$, $I_{C5} = 2.16\,\text{mA}$
all transistors active, all base currents negligible.

Solution 4B.15 **Hard**



Step 1 — Assumptions

Assume every transistor is in the active region and that every base current is negligible ($I_B \approx 0$, tool T1): then $|V_{BE}| = 0.7\,\text{V}$, $I_C \approx I_E$, and no current flows into any base, so a base network behaves like a plain resistor divider.

Step 2 — Walk through the circuit

1. Q_1 (diode-connected) is active: $V_{B1} = 6 - 0.7 = 5.3\,\text{V}$.
2. Q_3 is pnp and assumed active: $V_{BE} = -0.7\,\text{V}$, so $V_{C2} = 0 + 0.7 = 0.7\,\text{V}$.
3. KCL at the collector of Q_1 (base current neglected):

$$I_{C1} = \frac{V_{B1} - (-6)}{R_1} = 2.13\,\text{mA}$$

4. Q_2 has the same base and emitter nodes as Q_1, hence the same V_{BE} and the same current (current mirror): $I_{C2} = I_{C1} = 2.13\,\text{mA}$.
5. KCL at the emitter of Q_3 (base current neglected):

$$I_{E3} = I_{C2} = 2.13\,\text{mA}, \qquad I_{C3} \approx I_{E3}$$

6. The current $I_{C3} = 2.13\,\text{mA}$ flows from V_{C3} through R_2: $V_{C3} = (-6) + R_2\,(I_{C3}) = -6 + 1.8\,\text{k} \times 2.13\,\text{mA} = -2.16\,\text{V}$.
7. Q_4 is npn and assumed active: $V_{BE} = 0.7\,\text{V}$, so $V_{E4} = V_{C3} - 0.7 = -2.86\,\text{V}$.
8. KCL at the emitter of Q_4 (base current neglected):

$$I_{E4} = \frac{V_{E4} - (-6)}{R_4} = 2.61\,\text{mA}, \qquad I_{C4} \approx I_{E4}$$

9. Q_4 draws $I_{C4} = 2.61\,\text{mA}$ from V_{C4} through R_3: $V_{C4} = 6 - R_3\,(I_{C4}) = 6 - 1\,\text{k} \times 2.61\,\text{mA} = 3.39\,\text{V}$.

Step 3 — Verify the assumptions

- $I_{B1} = \frac{I_{C1}}{\beta} = 21.3\,\mu\text{A}$ (all bases on this node: $42.6\,\mu\text{A}$) $\ll$ the $2.13\,\text{mA}$ flowing in the base network ✓
- $I_{B2} = \frac{I_{C2}}{\beta} = 21.3\,\mu\text{A}$ (all bases on this node: $42.6\,\mu\text{A}$) $\ll$ the $2.13\,\text{mA}$ flowing in the base network ✓
- $I_{B4} = \frac{I_{C4}}{\beta} = 26.1\,\mu\text{A} \ll$ the $2.13\,\text{mA}$ flowing in the base network ✓
- $V_{EC1} = 6 - V_{B1} = 6 - (5.3) = 0.7\,\text{V} \geq 0.2\,\text{V} \Rightarrow Q_1$ is active ✓
- $V_{EC2} = 6 - V_{C2} = 6 - (0.7) = 5.3\,\text{V} \geq 0.2\,\text{V} \Rightarrow Q_2$ is active ✓
- $V_{EC3} = V_{C2} - V_{C3} = 0.7 - (-2.162) = 2.86\,\text{V} \geq 0.2\,\text{V} \Rightarrow Q_3$ is active ✓
- $V_{CE4} = V_{C4} - V_{E4} = 3.385 - (-2.862) = 6.25\,\text{V} \geq 0.2\,\text{V} \Rightarrow Q_4$ is active ✓

Step 4 — How good was $I_B \approx 0$?

If we keep the base currents ($I_C = \beta I_B$ with $\beta = 100$) and solve the circuit exactly, the currents change only a little:

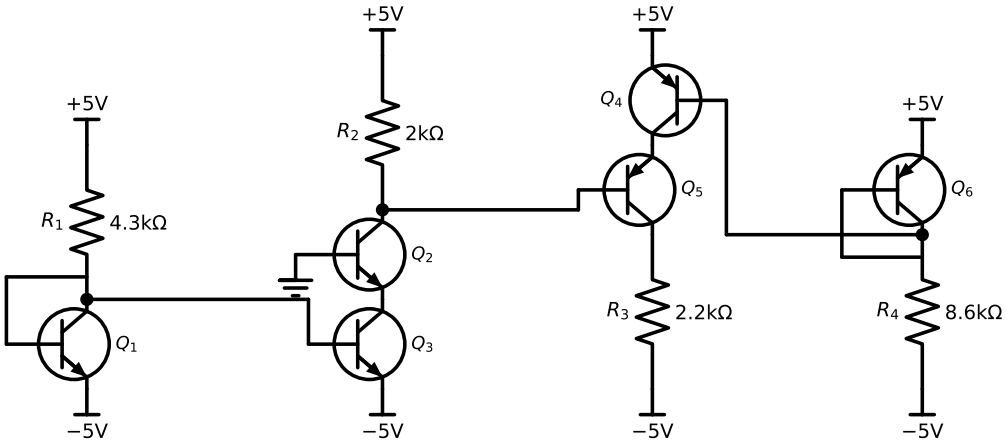
transistor	I_C (with $I_B \approx 0$)	I_C (exact)	difference
Q_1	$2.13\,\text{mA}$	$2.09\,\text{mA}$	2%
Q_2	$2.13\,\text{mA}$	$2.09\,\text{mA}$	2%
Q_3	$2.13\,\text{mA}$	$2.07\,\text{mA}$	3%
Q_4	$2.61\,\text{mA}$	$2.46\,\text{mA}$	6.3%

The differences grow along the chain: every base current slightly shifts the collector voltage that biases the next transistor, and $\alpha = \beta/(\beta+1)$ makes each I_C about 1 % smaller than its I_E. A shift is amplified when the next stage has only a small voltage across its emitter resistor. Hand analysis with $I_B \approx 0$ is still the standard first pass.

Answer

$I_{C1} = 2.13\,\text{mA}, \quad I_{C2} = 2.13\,\text{mA}, \quad I_{C3} = 2.13\,\text{mA}, \quad I_{C4} = 2.61\,\text{mA}$
all transistors active, all base currents negligible.

Solution 4B.16 **Hard**



Step 1 — Assumptions

Assume every transistor is in the active region and that every base current is negligible ($I_B \approx 0$, tool T1): then $|V_{BE}| = 0.7\,\text{V}$, $I_C \approx I_E$, and no current flows into any base, so a base network behaves like a plain resistor divider.

Step 2 — Walk through the circuit

1. Q_1 (diode-connected) is active: $V_{B1} = (-5) + 0.7 = -4.3\,\text{V}$.
2. Q_2 is npn and assumed active: $V_{BE} = 0.7\,\text{V}$, so $V_{E2} = 0 - 0.7 = -0.7\,\text{V}$.
3. Q_4 is active: $V_{B4} = 5 - 0.7 = 4.3\,\text{V}$.
4. KCL at the collector of Q_1 (base current neglected):

$$I_{C1} = \frac{5 - V_{B1}}{R_1} = 2.16\,\text{mA}$$

5. KCL at the collector of Q_6 (base current neglected):

$$I_{C6} = \frac{V_{B4} - (-5)}{R_4} = 1.08\,\text{mA}$$

6. Q_3 has the same base and emitter nodes as Q_1, hence the same V_{BE} and the same current (current mirror): $I_{C3} = I_{C1} = 2.16\,\text{mA}$.
7. Q_4 has the same base and emitter nodes as Q_6, hence the same V_{BE} and the same current (current mirror): $I_{C4} = I_{C6} = 1.08\,\text{mA}$.
8. KCL at the emitter of Q_2 (base current neglected):

$$I_{E2} = I_{C3} = 2.16\,\text{mA}, \qquad I_{C2} \approx I_{E2}$$

9. Q_2 draws $I_{C2} = 2.16\,\text{mA}$ from V_{C2} through R_2: $V_{C2} = 5 - R_2\,(I_{C2}) = 5 - 2\,\text{k} \times 2.16\,\text{mA} = 0.674\,\text{V}$.
10. Q_5 is pnp and assumed active: $V_{BE} = -0.7\,\text{V}$, so $V_{C4} = V_{C2} + 0.7 = 1.37\,\text{V}$.

11. KCL at the emitter of Q_5 (base current neglected):

$$I_{E5} = I_{C4} = 1.08\,\text{mA}, \qquad I_{C5} \approx I_{E5}$$

12. The current $I_{C5} = 1.08\,\text{mA}$ flows from V_{C5} through R_3: $V_{C5} = (-5) + R_3\,(I_{C5}) = -5 + 2.2\,\text{k} \times 1.08\,\text{mA} = -2.62\,\text{V}$.

Step 3 — Verify the assumptions

- $I_{B1} = \frac{I_{C1}}{\beta} = 21.6\,\mu\text{A}$ (all bases on this node: $43.3\,\mu\text{A}$) $\ll$ the $2.16\,\text{mA}$ flowing in the base network ✓
- $I_{B3} = \frac{I_{C3}}{\beta} = 21.6\,\mu\text{A}$ (all bases on this node: $43.3\,\mu\text{A}$) $\ll$ the $2.16\,\text{mA}$ flowing in the base network ✓
- $I_{B4} = \frac{I_{C4}}{\beta} = 10.8\,\mu\text{A}$ (all bases on this node: $21.6\,\mu\text{A}$) $\ll$ the $1.08\,\text{mA}$ flowing in the base network ✓
- $I_{B5} = \frac{I_{C5}}{\beta} = 10.8\,\mu\text{A} \ll$ the $2.16\,\text{mA}$ flowing in the base network ✓
- $I_{B6} = \frac{I_{C6}}{\beta} = 10.8\,\mu\text{A}$ (all bases on this node: $21.6\,\mu\text{A}$) $\ll$ the $1.08\,\text{mA}$ flowing in the base network ✓
- $V_{CE1} = V_{B1} - (-5) = -4.3 - (-5) = 0.7\,\text{V} \geq 0.2\,\text{V} \Rightarrow Q_1$ is active ✓
- $V_{CE2} = V_{C2} - V_{E2} = 0.6744 - (-0.7) = 1.37\,\text{V} \geq 0.2\,\text{V} \Rightarrow Q_2$ is active ✓
- $V_{CE3} = V_{E2} - (-5) = -0.7 - (-5) = 4.3\,\text{V} \geq 0.2\,\text{V} \Rightarrow Q_3$ is active ✓
- $V_{EC4} = 5 - V_{C4} = 5 - (1.374) = 3.63\,\text{V} \geq 0.2\,\text{V} \Rightarrow Q_4$ is active ✓
- $V_{EC5} = V_{C4} - V_{C5} = 1.374 - (-2.621) = 4\,\text{V} \geq 0.2\,\text{V} \Rightarrow Q_5$ is active ✓
- $V_{EC6} = 5 - V_{B4} = 5 - (4.3) = 0.7\,\text{V} \geq 0.2\,\text{V} \Rightarrow Q_6$ is active ✓

Step 4 — How good was $I_B \approx 0$?

If we keep the base currents ($I_C = \beta I_B$ with $\beta = 100$) and solve the circuit exactly, the currents change only a little:

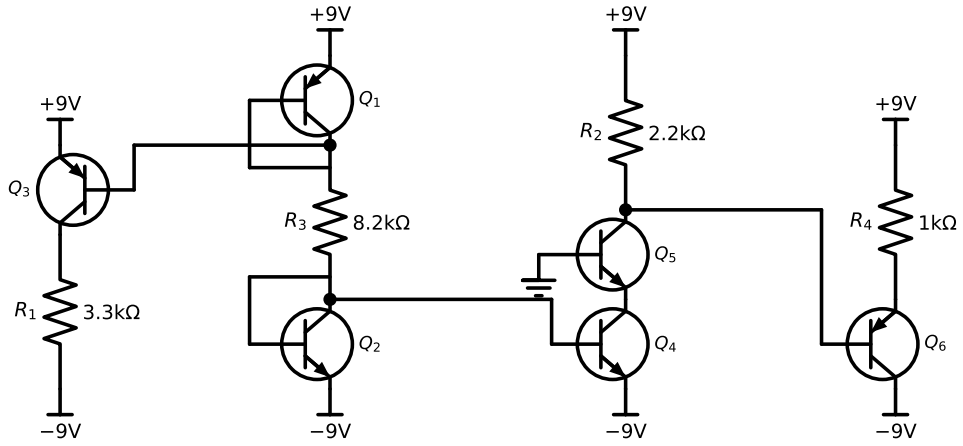
transistor	I_C (with $I_B \approx 0$)	I_C (exact)	difference
Q_1	2.16 mA	2.12 mA	2%
Q_2	2.16 mA	2.1 mA	3%
Q_3	2.16 mA	2.12 mA	2%
Q_4	1.08 mA	1.06 mA	2%
Q_5	1.08 mA	1.05 mA	3%
Q_6	1.08 mA	1.06 mA	2%

Answer

$I_{C1} = 2.16\,\text{mA}$, $I_{C2} = 2.16\,\text{mA}$, $I_{C3} = 2.16\,\text{mA}$, $I_{C4} = 1.08\,\text{mA}$, $I_{C5} = 1.08\,\text{mA}$, $I_{C6} = 1.08\,\text{mA}$

all transistors active, all base currents negligible.

Solution 4B.17 **Hard**



Step 1 — Assumptions

Assume every transistor is in the active region and that every base current is negligible ($I_B \approx 0$, tool T1): then $|V_{BE}| = 0.7\,\text{V}$, $I_C \approx I_E$, and no current flows into any base, so a base network behaves like a plain resistor divider.

Step 2 — Walk through the circuit

1. Q_3 is active: $V_{B3} = 9 - 0.7 = 8.3\,\text{V}$.
2. Q_2 (diode-connected) is active: $V_{B2} = (-9) + 0.7 = -8.3\,\text{V}$.
3. Q_5 is npn and assumed active: $V_{BE} = 0.7\,\text{V}$, so $V_{E5} = 0 - 0.7 = -0.7\,\text{V}$.
4. KCL at the collector of Q_2 (base current neglected):

$$I_{C2} = \frac{V_{B3} - V_{B2}}{R_3} = 2.02\,\text{mA}$$

5. KCL at the collector of Q_1 (base current neglected):

$$I_{C1} = \frac{V_{B3} - V_{B2}}{R_3} = 2.02\,\text{mA}$$

6. Q_3 has the same base and emitter nodes as Q_1, hence the same V_{BE} and the same current (current mirror): $I_{C3} = I_{C1} = 2.02\,\text{mA}$.
7. Q_4 has the same base and emitter nodes as Q_2, hence the same V_{BE} and the same current (current mirror): $I_{C4} = I_{C2} = 2.02\,\text{mA}$.
8. The current $I_{C3} = 2.02\,\text{mA}$ flows from V_{C3} through R_1: $V_{C3} = (-9) + R_1\,(I_{C3}) = -9 + 3.3\,\text{k} \times 2.02\,\text{mA} = -2.32\,\text{V}$.
9. KCL at the emitter of Q_5 (base current neglected):

$$I_{E5} = I_{C4} = 2.02\,\text{mA}, \qquad I_{C5} \approx I_{E5}$$

10. Q_5 draws $I_{C5} = 2.02\,\text{mA}$ from V_{C5} through R_2: $V_{C5} = 9 - R_2\,(I_{C5}) = 9 - 2.2\,\text{k} \times 2.02\,\text{mA} = 4.55\,\text{V}$.

11. Q_6 is pnp and assumed active: $V_{BE} = -0.7\,\text{V}$, so $V_{E6} = V_{C5} + 0.7 = 5.25\,\text{V}$.

12. KCL at the emitter of Q_6 (base current neglected):

$$I_{E6} = \frac{9 - V_{E6}}{R_4} = 3.75\,\text{mA}, \qquad I_{C6} \approx I_{E6}$$

Step 3 — Verify the assumptions

- $I_{B3} = \frac{I_{C3}}{\beta} = 20.2\,\mu\text{A}$ (all bases on this node: $40.5\,\mu\text{A}$) $\ll$ the $2.02\,\text{mA}$ flowing in the base network ✓
- $I_{B1} = \frac{I_{C1}}{\beta} = 20.2\,\mu\text{A}$ (all bases on this node: $40.5\,\mu\text{A}$) $\ll$ the $2.02\,\text{mA}$ flowing in the base network ✓
- $I_{B2} = \frac{I_{C2}}{\beta} = 20.2\,\mu\text{A}$ (all bases on this node: $40.5\,\mu\text{A}$) $\ll$ the $2.02\,\text{mA}$ flowing in the base network ✓
- $I_{B4} = \frac{I_{C4}}{\beta} = 20.2\,\mu\text{A}$ (all bases on this node: $40.5\,\mu\text{A}$) $\ll$ the $2.02\,\text{mA}$ flowing in the base network ✓
- $I_{B6} = \frac{I_{C6}}{\beta} = 37.5\,\mu\text{A} \ll$ the $2.02\,\text{mA}$ flowing in the base network ✓
- $V_{EC3} = 9 - V_{C3} = 9 - (-2.32) = 11.3\,\text{V} \geq 0.2\,\text{V} \Rightarrow Q_3$ is active ✓
- $V_{EC1} = 9 - V_{B3} = 9 - (8.3) = 0.7\,\text{V} \geq 0.2\,\text{V} \Rightarrow Q_1$ is active ✓
- $V_{CE2} = V_{B2} - (-9) = -8.3 - (-9) = 0.7\,\text{V} \geq 0.2\,\text{V} \Rightarrow Q_2$ is active ✓
- $V_{CE5} = V_{C5} - V_{E5} = 4.546 - (-0.7) = 5.25\,\text{V} \geq 0.2\,\text{V} \Rightarrow Q_5$ is active ✓
- $V_{CE4} = V_{E5} - (-9) = -0.7 - (-9) = 8.3\,\text{V} \geq 0.2\,\text{V} \Rightarrow Q_4$ is active ✓
- $V_{EC6} = V_{E6} - (-9) = 5.246 - (-9) = 14.2\,\text{V} \geq 0.2\,\text{V} \Rightarrow Q_6$ is active ✓

Step 4 — How good was $I_B \approx 0$?

If we keep the base currents ($I_C = \beta I_B$ with $\beta = 100$) and solve the circuit exactly, the currents change only a little:

transistor	I_C (with $I_B \approx 0$)	I_C (exact)	difference
Q_3	2.02 mA	1.98 mA	2%
Q_1	2.02 mA	1.98 mA	2%
Q_2	2.02 mA	1.98 mA	2%
Q_5	2.02 mA	1.97 mA	3%
Q_4	2.02 mA	1.98 mA	2%
Q_6	3.75 mA	3.51 mA	6.9%

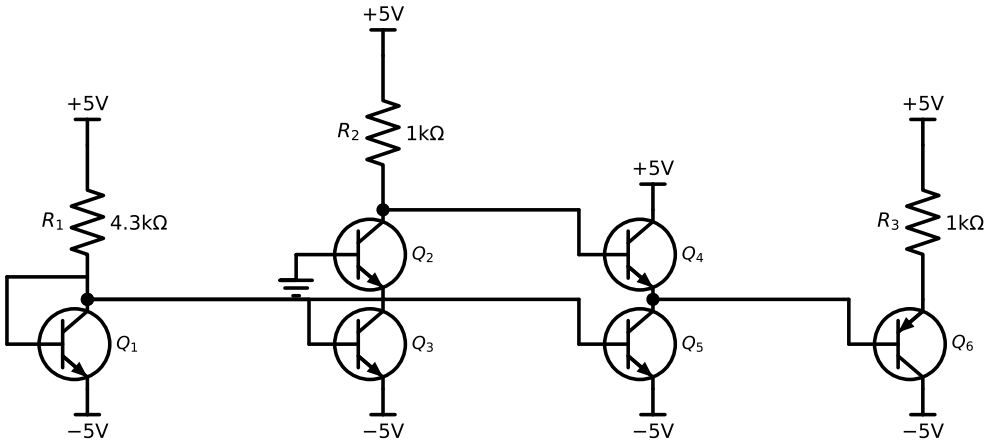
The differences grow along the chain: every base current slightly shifts the collector voltage that biases the next transistor, and $\alpha = \beta/(\beta+1)$ makes each I_C about 1 % smaller than its I_E. A shift is amplified when the next stage has only a small voltage across its emitter resistor. Hand analysis with $I_B \approx 0$ is still the standard first pass.

Answer

$I_{C3} = 2.02\,\text{mA}$, $I_{C1} = 2.02\,\text{mA}$, $I_{C2} = 2.02\,\text{mA}$, $I_{C5} = 2.02\,\text{mA}$, $I_{C4} = 2.02\,\text{mA}$, $I_{C6} = 3.75\,\text{mA}$

all transistors active, all base currents negligible.

Solution 4B.18 **Hard**



Step 1 — Assumptions

Assume every transistor is in the active region and that every base current is negligible ($I_B \approx 0$, tool T1): then $|V_{BE}| = 0.7\,\text{V}$, $I_C \approx I_E$, and no current flows into any base, so a base network behaves like a plain resistor divider.

Step 2 — Walk through the circuit

1. Q_1 (diode-connected) is active: $V_{B1} = (-5) + 0.7 = -4.3\,\text{V}$.
2. Q_2 is npn and assumed active: $V_{BE} = 0.7\,\text{V}$, so $V_{E2} = 0 - 0.7 = -0.7\,\text{V}$.
3. KCL at the collector of Q_1 (base current neglected):

$$I_{C1} = \frac{5 - V_{B1}}{R_1} = 2.16\,\text{mA}$$

4. Q_3 has the same base and emitter nodes as Q_1, hence the same V_{BE} and the same current (current mirror): $I_{C3} = I_{C1} = 2.16\,\text{mA}$.
5. Q_5 has the same base and emitter nodes as Q_1, hence the same V_{BE} and the same current (current mirror): $I_{C5} = I_{C1} = 2.16\,\text{mA}$.
6. KCL at the emitter of Q_2 (base current neglected):

$$I_{E2} = I_{C3} = 2.16\,\text{mA}, \qquad I_{C2} \approx I_{E2}$$

7. Q_2 draws $I_{C2} = 2.16\,\text{mA}$ from V_{C2} through R_2: $V_{C2} = 5 - R_2\,(I_{C2}) = 5 - 1\,\text{k} \times 2.16\,\text{mA} = 2.84\,\text{V}$.
8. Q_4 is npn and assumed active: $V_{BE} = 0.7\,\text{V}$, so $V_{E4} = V_{C2} - 0.7 = 2.14\,\text{V}$.
9. Q_6 is pnp and assumed active: $V_{BE} = -0.7\,\text{V}$, so $V_{E6} = V_{E4} + 0.7 = 2.84\,\text{V}$.
10. KCL at the emitter of Q_4 (base current neglected):

$$I_{E4} = I_{C5} = 2.16\,\text{mA}, \qquad I_{C4} \approx I_{E4}$$

11. KCL at the emitter of Q_6 (base current neglected):

$$I_{E6} = \frac{5 - V_{E6}}{R_3} = 2.16\,\text{mA}, \qquad I_{C6} \approx I_{E6}$$

Step 3 — Verify the assumptions

- $I_{B1} = \frac{I_{C1}}{\beta} = 21.6\,\mu\text{A}$ (all bases on this node: $64.9\,\mu\text{A}$) $\ll$ the $2.16\,\text{mA}$ flowing in the base network ✓
- $I_{B3} = \frac{I_{C3}}{\beta} = 21.6\,\mu\text{A}$ (all bases on this node: $64.9\,\mu\text{A}$) $\ll$ the $2.16\,\text{mA}$ flowing in the base network ✓
- $I_{B4} = \frac{I_{C4}}{\beta} = 21.6\,\mu\text{A} \ll$ the $2.16\,\text{mA}$ flowing in the base network ✓
- $I_{B5} = \frac{I_{C5}}{\beta} = 21.6\,\mu\text{A}$ (all bases on this node: $64.9\,\mu\text{A}$) $\ll$ the $2.16\,\text{mA}$ flowing in the base network ✓
- $V_{CE1} = V_{B1} - (-5) = -4.3 - (-5) = 0.7\,\text{V} \geq 0.2\,\text{V} \Rightarrow Q_1$ is active ✓
- $V_{CE2} = V_{C2} - V_{E2} = 2.837 - (-0.7) = 3.54\,\text{V} \geq 0.2\,\text{V} \Rightarrow Q_2$ is active ✓
- $V_{CE3} = V_{E2} - (-5) = -0.7 - (-5) = 4.3\,\text{V} \geq 0.2\,\text{V} \Rightarrow Q_3$ is active ✓
- $V_{CE4} = 5 - V_{E4} = 5 - (2.137) = 2.86\,\text{V} \geq 0.2\,\text{V} \Rightarrow Q_4$ is active ✓
- $V_{CE5} = V_{E4} - (-5) = 2.137 - (-5) = 7.14\,\text{V} \geq 0.2\,\text{V} \Rightarrow Q_5$ is active ✓
- $V_{EC6} = V_{E6} - (-5) = 2.837 - (-5) = 7.84\,\text{V} \geq 0.2\,\text{V} \Rightarrow Q_6$ is active ✓

Step 4 — How good was $I_B \approx 0$?

If we keep the base currents ($I_C = \beta I_B$ with $\beta = 100$) and solve the circuit exactly, the currents change only a little:

transistor	I_C (with $I_B \approx 0$)	I_C (exact)	difference
Q_1	$2.16\,\text{mA}$	$2.1\,\text{mA}$	3%
Q_2	$2.16\,\text{mA}$	$2.08\,\text{mA}$	4%
Q_3	$2.16\,\text{mA}$	$2.1\,\text{mA}$	3%
Q_4	$2.16\,\text{mA}$	$2.06\,\text{mA}$	5.1%
Q_5	$2.16\,\text{mA}$	$2.1\,\text{mA}$	3%
Q_6	$2.16\,\text{mA}$	$2.08\,\text{mA}$	4%

Answer

$I_{C1} = 2.16\,\text{mA}$, $I_{C2} = 2.16\,\text{mA}$, $I_{C3} = 2.16\,\text{mA}$, $I_{C4} = 2.16\,\text{mA}$, $I_{C5} = 2.16\,\text{mA}$, $I_{C6} = 2.16\,\text{mA}$

all transistors active, all base currents negligible.